Engineering Mechanics 1 · Unit_02 · UNIT 2

DRAFT — UE26ME142A · Unit_02 — UNIT 2 · Complete Unit Master

Notes-first study master · source-cited lecture explanations and compact page takeaways

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the edited lesson notes by topic. Evidence chips point to transcript intervals and slide pages. The complete source transcripts, page text, formula candidates, and image locators are preserved in sibling JSON datasets. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Unit_02 · UNIT 2

1 UE26ME142A · Unit_02: UNIT 2

This unit master organizes source material for Engineering Mechanics 1, Unit_02, officially titled UNIT 2. It includes 15 video transcripts and 18 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Unit map · UE26ME142A · Unit_02: UNIT 2

Official catalogue entry: UE26ME142A, Unit_02 (UNIT 2). Bucket folder present: yes. Videos: 15; slide decks: 18.

EV0000001

2 UE26ME142A_Unit 2_Class 14_AVSummary

Study focus: UE26ME142A_Unit 2_Class 14_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- SV Venkatish from Civil Engineering and the Department of Science and Humanities PS University.
- Two chapters which are there in unit one, chapter one and chapter two.
- In chapter one, we had discussed a few of the definitions and the units of forces.
- In chapter two, we had four topics.

- One as resolution of forces.
- In that you have rectangular components, N and T components, then axes which are not at 90 degrees, then the difference between projection and components.
- That's topic one under chapter two.
- The second topic was on movement.
- We discussed moment, varingance principle, and we have solved problems on moment also.
- That's second topic.
- The third topic was on couple.
- We have seen what is the definition of couple, characteristics of couple, classification of couple, effects of couple, and very importantly, what do you mean by force couple system?
- And then we have solved problems on that.
- The last topic in unit one, chapter two, is resulted.
- We have discussed what is resulted, how do we determine the characteristics of resultant, four characteristics that is magnitude, direction, sense, and point of application.
- To determine the point of application, we have used principle of moments, okay, and then we have solved problems on that.
- To certain extent unit 3 also.
- You have the background of unit one.
- We will now get into unit two.
- And the prescribed textbook that we are using is engineering mechanics statics.
- Very importantly, it is the eighth edition, which is in SI version.
- And the author is Miriam Craig and adopted by Bolton.
- There's one more in this other than equilibrium, we also have trusses.
- We will discuss that later when we take up that topic.
- So first we will discuss about chapter three, which is equilibrium.
- What is equilibrium?
- If the resultant force and the resultant movement acting on a body on which there are several.
- Forces and moments acting, that is a system of forces are acting.
- If the resultant force and moment both of them are zero, then we say that the body is in equilibrium.
- Application of the force would be a rotatory motion which will make this body to rotate.
- It won't move.
- But you can also have a combination of transmitter motion as well as rotatory motion.
- It ensures that there is no transmitter emotion.
- Why I'm saying this?
- Because as you could see, the net force because of a couple is zero.
- But what is the effect of a couple?
- It makes the body to rotate.
- It should not get displaced, it should not rotate also.
- But if we take this, m is equal to 0, or algebraic sum of all the moments of all the forces at any point, at any point should be equal to 0.
- This will ensure that there is no rotatory motion.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

Ensure or to show that the body is in equilibrium or to determine the forces which makes the body to be in equilibrium.

Key details

- What is that?
- The first thing is a free body diagram.
- We have already discussed about free body diagram.
- But since it comes exclusively in this chapter, I am again arbitrating or bringing the concept of a free body diagram.
- As the name suggests, free body diagram is a diagram in which we will have a diagram.
- Such that there is no body in that.
- The diagram will have the forces, the characteristics of forces which are acting on the body, and we were going to remove that body because we are not interested in the body.
- What we are interested is what are the forces that is acting on the body, what is the effect of that.
- That is what is called as a free body diagram, as I suggest, as the name itself suggests, we are making the diagram free of the body.
- Position means the point of application.
- In the solution of problems in mechanics, if how to draw the free body diagram, fifty percent of the problem is solved when it comes to equilibrium chapter.
- But that free body diagram should be clear and sufficient for you to determine the unknowns.
- A few of these supports we have already discussed in our unit one.
- The first type of support is a flexible cable or belt or chain or rope.
- It is a body which can support tensile forces.
- When I say tensile forces, if you have a rope and if you try to pull it, okay, once it is tight, afterwards it will not move.
- For this type of support where you have a flexible cable or a bolt, or sorry, belt, or a chain, or a rope.
- It can resist tensile forces, and that is what is indicated here.
- So this, if I try to pull this it will not move because of this oh.
- And there's a tension here, the internal forces will be acting towards each other and that is what is indicated here.
- If I cut it here and look at this, I will get the force to be this.
- There will be only one force with no line of action.
- If you are looking at what is the force acting on the in the row, then it is internal.
- If you are looking at what is the force causing the tension is the external force.
- The second one is a smooth surface.
- A smooth surface is one in which you will have one reaction with known line of action.
- Which will always act perpendicular to the surface.

- This is the surface.
- And this is the reaction that we get and this angle is always ninety degrees and the sense is going to be opposite to the body weight that is acting.
- The body weight will be acting here, so you will have the normal reaction acting here.
- There won't be any friction when it comes to smooth surface.
- Or the frictional force that develops because of the smooth surface would be quite less when compared to the force that is acting.
- The next category is a rough surface.
- You have already studied about this.
- If the roughness of the body which are in contact with each other is quite large.
- Then it will prevent the body from moving tangential to the surface, and that we call it as frictional force.
- In addition to this frictional force, we will also have a normal reaction.

Example or application from the lecture

- Let us look at that.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

Core explanation · 10:00–14:59

to the normal.

Key details

- We can represent these two by one resultant force, but the direction of this resultant force depends upon the magnitude of F1N.
- The next category of the type of support is a roller support.
- Four different ways of representing the rollers.
- You can also have a roller vertical or inclined also.
- with known line of action, which will always be acting perpendicular to the line of rollers.
- In this case the line of roller is horizontal in all the four cases.
- You can also have vertical if the rollers are to the left, then you will have the support reaction acting to the right.
- If you have the rollers.
- To the right, then you will have the reaction to the left.
- The next one is a freely sliding guide.
- In this case, you will have one reaction only.
- In this case, you can see that the sliding guide can move horizontally, it cannot move vertically.
- The reaction can be upwards or downwards.
- Along the horizontal, it can move only this much.
- After which, if it moves till here, then it cannot move.
- Then you will have a reaction in the horizontal direction also.
- That is why it is called as freely sliding train.

- Only till that position.
- The horizontal support will not be there.
- After that, you will have the horizontal support.
- But however, here what is indicated for this position is there will be only one normal reaction, normal reacting vertically.
- Continuing the different types of supports, the next one comes.
- Pin connection, look at how the pin connection is indicated.
- There is a clear difference between a pin connection and a roller connection.
- In case of rollers, you will have roller here and then you will have a the surface.
- Whereas pin connection, directly the triangle is resting on the surface.
- The door frame or the door itself is fixed to the frame of the door.
- In case of hinges, you will have two reaction.
- No, you cannot move the door upwards or downwards.
- Toward the frame, or you cannot pull it out of the frame, but you can rotate the door.
- That is permitted in case of the door.
- That is what the hint support does.
- It will provide you two reactions.
- It will permit you to rotate.
- In the plane opposite to this two reaction that is coming.
- If this is X and Y, it will rotate along the Z axis.
- No, you have a door with a stopper.
- Doe stopper, we are we know the door stopper.
- Because the doorstopper, it will prevent the door from closing.
- Then we will have the moment also.
- That is why you can see here.
- Pin not free to turn.
- Where you will have restriction for rotation.
- The next one is a fixed or a built-end support.
- This is welded connection, this is fixed.
- The sense of this, even here the sense of this reactions can be upwards or downwards, right side or left side, clockwise or anti-clockwise based on the force that is acting on the body.
- Then we have also looked at gravitational force earlier.
- Always the gravitational force will be acting downwards.

Example or application from the lecture

- The best example that I can give you is the door.
- Hinges are the best example for pin connection.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

Core explanation · 15:00–19:59

Okay, you cannot round it off to ten because if you round it off to ten then you will not get the correct answer what is mentioned in the textbook.

Key details

- Next other one important type of support is a spring connection.
- The spring, in case of spring connection, the spring will allow it to displace or it will allow it to elongate or compress, but with certain force.
- Which defines what should be the force that you need to apply for unit elongation or compression.
- That is what is called as stiffness K.
- elongation or compression that I have to give, then that we can find it as F is equal to K times the elongation or compression.
- Along with this, we also have one more, what is called as torsional spring action, where instead of a linear spring, this is a linear spring where you can elongate or compress.
- In case of torsional spring,.
- You can you have to apply a movement so that it rotates.
- That is m by theta.
- Yes, KS or KT torsional spring is equal to what is the moment required for an unit displacement.
- You can see here the body as such is removed, only the outer edges are intact.
- Then the supports replaced by their respective reactions.
- Here it is an inch support, so you have two reactions, here it is a roller support, so you have one reaction.
- If you don't understand, you can always come and talk to me later.
- First one now here you can see there is a force acting here.
- This is a cable, so there will be one reaction and they have indicated that tension.
- Here they have kept a mass here.
- So because of this mass there will be a force which is m times g.
- Here you have a.
- Hinge support which is not indicated here.
- You have to identify what is incomplete here.
- And according to that you should complete the free body diagram.
- I hope you can do that.
- Next, this table gives you a free body diagram, which is wrong or incomplete.
- You can see here it is wrong or incomplete.
- I'll explain the first one.
- You have a roller inclined on an inclined surface, or kept on an inclined surface, there's a force of P that is applied.
- What is wrong?
- The force P is acting here, that is exactly right, P is correct.
- If this is a smooth surface, so you have one reaction normal to the surface, that is also correct.
- Mg is the force because of the sulfate of the roller.
- That is because of gravitation.
- It should act vertically down.
- You have to draw the free body diagram for all this, sorry, eight diagrams, all this eight diagrams.

- I will explain the first one.

Example or application from the lecture

- But you can see here they have given Mg.

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EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

Core explanation · 20:00–24:59

Suspended by vertical cable at A and supported by rough inclined surface at B.

Key details

- This is a cable.
- Then this I will remove it and replace it by a force acting upwards because it's tension, the internal forces will be acting upwards.
- Whereas here there will be a normal reaction acting upwards, then there will be a frictional force acting along this surface and it is upwards.
- If you have any doubts you can meet.
- We will stop it here.
- We'll continue in the next class.
- My name is Dr.
- Sv Vength from Civil Engineering under Department of Science and Humanities, PS University.
- This is class number eighteen.
- Next we will have class number nineteen.

Example or application from the lecture

- Let's meet in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

3 UE26ME142A_Unit 2_Class 15_AVSummary

Study focus: UE26ME142A_Unit 2_Class 15_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000022 · EV0000023

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Svend Kitish from Civil Engineering under Department of Science and Humanities, PS University.
- This is class number nineteen.
- And this is class number two.
- Under unit two.
- Under unit two, in class one, that is class number eighteen, we had discussed about the definition of equilibrium.
- What are the equity that need to be considered for a body to be in equilibrium?
- Then we have looked at few of the concepts or definitions, or what is that is required for us to know to understand equilibrium.
- How do we draw a free body diagram?
- And various different.
- What are the different types of supports that we should know?
- And what are the reactions that is going to be acting because of those supports were written in the first class of unit 2.
- Today we will be continuing in this class further.
- The first one is collinear.
- In this case, it is taken that the forces is acting along X axis.
- If this equation is satisfied, then we can easily say.
- That the body is in equilibrium.
- We don't have to look at anything else.
- Because there is no force acting perpendicular to it.
- Then movement about any point is equal to zero, because if the net force acting is zero.
- Then that would not produce any movement except if it is a couple.
- But in this case, since all the forces are acting along the same axis, that would not produce any couple.
- I hope that is clear.
- To specify whether the body is in equilibrium or not.
- If the algebra sum of all the forces is equal to zero, the body will be in equilibrium.
- The second category of the force system is concurrent at a point.
- What do you mean by concurrent?
- The point of application of all the forces when extended would meet at the same point.
- Like this, whatever is indicated.
- And if I resolve these forces along this axis X and Y, and if I could get the algebra sum of all the forces along X to be equal to 0, and the algebraic sum of all the forces along y to be equal to 0.
- That means the result is 0.
- If this is proved, then I don't have to go and take the other equation that is sigma m equals to 0.
- The body is in equilibrium.
- That is because the point of application of all these forces meet at the same point O.
- But the resultant is equal to zero because summation of all the forces along X is 0, along Y is also 0.

- That is only when there is concurrent, the forces are concurrent.
- The next category is a parallel force system.
- There are four forces acting.

Example or application from the lecture

- For example, a free body diagram.
- Tables were given where how to draw the free body diagram was explained and there were some diagrams which was explained which was wrong and how do we correct that also was explained.
- For example, summation of all the forces perpendicular to x-axis.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

Core explanation · 05:00–09:59

Yes, the displacement may not be there, but it can produce a moment.

Key details

- The forces or the summation of the moment of all the forces about the z-axis about any point should be equal to zero.
- If that is also satisfied, then this body would be in equilibrium.
- The next category is a general category where you have several force system.
- You can also have an applied moment.
- The forces when it is extended may not meet at the same point.
- In such case, we have to have all these three equations of equilibrium.
- Which should be satisfied, that is algebraic sum of all the forces acting along yes.
- Algebic sum of all the forces acting along Y?
- And the moment about the z-axis about any point should be equal to zero.
- If it satisfies all these three equations, then we can say that this body is in equilibrium.
- the component of all the forces about the algebraic sum of the components of all the forces about x axis should be zero.
- Then it means that there is no displacement along X.
- The algebraic sum of all the components of all the forces about y axis is 0, then I can also say that the displacement along y is 0.
- And if the movement of all the forces and the couple or the moment.
- About any point along the z axis is equal to zero, then there is no rotation.
- And these three equations are called as the equations of equilibrium for a body which is considered to be a two dimensional.
- We are not looking at three dimensional.
- If it is three dimensional, we will have additional equations that needs to be proved or additional things that needs to be proved.
- In addition to all these three, we will have another three equations.
- That is algebraic sum of the components of all the forces about the z axis also should be equal to zero.

- Then algebraic sum of the moments about x axis should be zero.
- Then algebraic sum of all the moments about the y axis also should be equal to zero.
- And six will be the equation of equilibrium for a three dimensional.
- But if it is two dimensional and if we take the body to be in X and Y axis, then these are the three equations of equilibrium, which has to be satisfied for us to say that the body is in equilibrium.
- Next continuation of this.
- If we have a two force members, how do we say whether the body is in equilibrium or not?
- If it is two force members, if it is collinear, and if one force is opposite to the other, plus.
- If the magnitude of the force is the same, then we can easily say that the body is in equilibrium.
- That is, if I take the algebraic sum of all the forces along this axis, okay, then I can easily say that the body is in equilibrium.
- Next, if I take a three force system, something like this, which is concurrent, then if this follows the law of triangle of forces as indicated here, then I can easily say that the body is in equilibrium.
- If you take the two forces on the adjacent sides of a parallelogram, if the other force forms the other side of the parallelogram, other side of the triangle taken in the opposite order, then we can easily say that the body is in equilibrium.
- Along this axis, and I also find that the movement about point A is zero.

Example or application from the lecture

- Say for example I take a body, there are several forces acting, and if I find that there is a result, if it exists, if there is a result and that is acting.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

Core explanation · 10:00–14:59

If we take an axis perpendicular to the resultant force if it exists, okay, and if the summation of all the forces along this axis is 0.

Key details

- And we also find that the moment about the point A, which is on this result and axis to be equal to zero.
- The movement about any point B where A and B is not perpendicular to R, if that is also equal to 0, then I can easily say that the body is in equilibrium.
- Other equations of equilibrium for such thing is your summation of all the forces about the x-axis should be equal to zero.
- The moment if a resultant exists.
- Along that line of action of the resultant, M A should be equal to 0.
- And also, if you take any point such that AB is not perpendicular to R, and if the movement about B is also equal to 0.
- Then I can easily say that the body is in equilibrium.

- Another condition that we take that if there exists a result, something like this, and if we found that the movement about point A is 0.
- And if I take some other point B and even there the moment because of all the forces is zero, that means that the resultant has to act along this A B line only.
- In addition to this, we take another point C where this AB or B C are not perpendicular to R, either this or this, and if we find that the moment of this about C is also 0.
- Then also we can say that the body is in equilibrium.
- So first thing is I have told you that sigma f x should be equal to 0, sigma f y should be equal to 0, and the sigma about the z axis about any point should be equal to 0.
- These are equations of equilibrium.
- Other than this, if I don't want to take, then I can also take this condition where I will be proving sigma fx equal to 0, sigma is equal to 0, where a is taken as the point along the line of action of the resultant if it exists.
- And also, if I can prove that sigma m b is equal to 0, where AB is not perpendicular to the resultant, if it exists, then I can say that the body is in equilibrium.
- Or if I take something like this, if there exists a resultant, and if I find that.
- About the point the moment is 0 where A is on the line of action of the resultant, and also if I take some other point B where AB is not perpendicular to R, and if I find that sigma M B is also 0, then your resultant if it takes it, it has to act along AB.
- And also I will take a point C where A C or A B C are not perpendicular to this resultant or this resultant.
- And if I find that movement about C because of this is also zero.
- Then I can easily say that the body is in equilibrium.
- This is these are the alternate equations of equipment that I can take.
- If I could resolve all the forces along X and Y, which I could do, then this constraint or this support will balance all the forces acting along the X axis, then there won't be any displacement along X.
- Then if the all the forces, the component of the force that is acting along y, if this support or constraint balances along with this of all this vertical force that is acting on the body, then there won't be any displacement along the y-axis.
- Now, if I take this point A, and if the moment about this point, because of all the forces that is acting along with this support reaction value.

Example or application from the lecture

- Consider here or the equations of equilibrium.
- Next, let's look at what are the fixity and adequate constraints that is required for me to say that the body is in equilibrium.
- Now there is an example that is taken here.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

Core explanation · 15:00–19:59

Will be equal to zero.

Key details

- That means there is no rotation that is going to occur.
- Along with the support reaction values.
- Then I can say that it is it has got a complete fixity and the constraint is also sufficient for complete fixity.
- Or for the equations of equilibrium to be valid.
- This may prevent the translatory motion along X, this may prevent the rotatory motion translatory motion along Y, but since this is acting along the line of action passes through A itself.
- Now moment about A because of this would not come into picture.
- So that means this body now can or able to rotate.
- Along this three, all this three is producing the displacement along X to be 0.
- Whereas when I apply a force here, vertical force, there is no constraint.
- It does not prevent it from displacing along Y.
- Or a partial constraint because it allows the displacement to occur a long way.
- Along with this, there is one more support here.
- Whether it passes through A or it does not pass through A, it does not matter.
- But if you compare this diagram and this diagram, only this support, this support and this support should have been enough for complete fixity and adequate constraint.
- But here we have excess fixity.
- One extra support is there.
- Whatever is required for equilibrium.
- More than that, we have provided a support.
- There is one redundant constraint that is existing.
- before we get into the problem solving mode.
- If it is excessive then you will have a redundant constraint.
- Then the sufficient should be sufficient so that there is no translatory motion or rotatory motion.
- If anyone is possible to occur, then we cannot say that the body is in equilibrium.
- And how do I how can how I can prove that?
- By using the or if it doesn't satisfy the equations of equilibrium, I can easily say that there is no adequate.
- constraint or I can say that it is partially constrained.
- We will continue in the next class that is class number 20.
- That would be the class number three under unit two.
- I am Doctor Sv Vengtesh.
- We will meet in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

4 UE26ME142A_Unit 2_Class 16_AVSummary

Study focus: UE26ME142A_Unit 2_Class 16_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000038 · EV0000039

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Svankitish, Civil Engineering, Department of Science and Humanities, PS University.
- This is class number twenty.
- in class number nineteen, that is previous class, we had discussed how do we go about writing the free body diagram.
- And then we have looked at what are the categories of equilibrium equations that we need to use for a particular type of force system.
- And also we discussed what are the alternate equations of equilibrium that we can use.
- to go out and solve different problems.
- Earlier to this, we had also looked at what are the different types of support that we can get and how do we convert this support or replace the support by their respective support reactions.
- Today we will be solving a few problems from the unsolved problems from the prescribed textbook that is Engineering Mechanics Statics authored by Miriam Craig and Bolton and published by Wiley Publications.
- That is 3 bar 12.
- The statement of the problem goes something like this.
- The 500 kg uniform beam is subjected to the three external loads shown.
- The XY plane is vertical.
- The self weight of the beam is 500 kgs, or I should say it is the mass of the beam is 500 kgs.
- And he says that the x-ray plane is vertical.
- You have a beam which is indicated by O to C and there is a support at O.
- If you recall this is a fixed support, then there are three forces, one is acting at point A, which is acting upwards, it is a point low.
- And the magnitude of that is 1.4 kilonewtons.
- Then we have a movement here which is in the clockwise or no sorry, anti-clockwise direction of magnitude 15 kilonewton meters acting at B.
- And the direction of this force is at thirty degrees to the vertical.
- The first thing that we need to do is to draw a free body diagram.
- Of course the free body diagram is not indicated here.
- But however I'm not be explaining how to draw the free body line.
- Replace the six set support by two reactions.
- Either I can take the sense to be acting upwards or downwards.

- Then you will have the horizontal reaction acting to the left or to the right.
- And then you will have a movement acting at war, which can be clockwise or indicwise.
- If you get a positive answer, the sense that you have adopted is correct.
- If you get a negative sign, that indicates that the sense that you have adopted should be opposite to what you have assumed.
- Then apply the force of 1.4 kilonot at A, apply the moment of 15 kilon meters at B, then you can resolve this force along vertical and horizontal.
- Then the horizontal component of this force will be three sin thirty degrees acting to the left.
- So once you draw the free body diagram, next use the equations of equilibrium.
- In equilibrium chapter, we have to draw the free body diagram.

Example or application from the lecture

- The distance of point A and B is also given, the horizontal distance from O to A is 1.2 meters, from A to B is 1.8 meters, and there is also an inclined point load of magnitude 3 kilonottons, acting at C.
- They have also given you the x axis and y axis.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

Core explanation · 05:00–09:59

Once you draw the free bad item, fifty percent of the problem will be solved.

Key details

- Then use the equations of equilibrium.
- When you are using the equations of equilibrium, remember that if you can use the movement equation first.
- And you choose the point where you want to take the moment as a point where there are unknown that needs to be determined.
- If there are two points where the unknown reactions are acting, then choose the point where the maximum unknown forces are acting as a point where you have to take the moment.
- By doing so, you will be able to get one of the unknown from this equight away.
- Then you can use the other two equation that is sigma f x is equal to 0 and sigma f y is equal to 0.
- So here first I have used sigma m not to be equal to 0 because that is the point where the support reactions are acting.
- Next, 15 kilonewton meters acting in the antipower direction, that would also be positive straight away, add that without multiplying any distance because that is the moment that is there.
- Or you can take the force vertical force that is acting because of this or the component of the 3 cannon force acting around the vertical, which will be 3 cos 30 into the total distance from O to C, which would be 1.2 plus 1.8 plus 1.8.
- Then the self weight of the beam, I forgot to tell you in the when I explained about the free body diagram.

- The self weight also has to be taken.
- The mass of the beam is 500 kgs.
- due to graph D which is G would be the force and that would be acting at the center of the beam which would be 1.2 plus 1.8 plus 1.8 divided by 2, which would be equal to 4.8 divided by 2, which will be 2.4.
- So this is the force that is caused because of the selfate which will be happening the center, that would create a clockwise movement, so it is negative into 4.8 is the total distance.
- The distance one 4.8 by 2.
- Now thousand is to convert the Newton, thus you will get a Newton because Kg into acceleration you to c you will get as 1 newton.
- And equate that to zero from this equation, we get the moment to be 7.56 kilon meters.
- You we have got as positive.
- Next, you take sigma f x equals 0, you have a reaction here.
- I'll assume the reaction at O to be acting to the right, but here the component of the three country force will be acting towards the left.
- And the component is negative, or I can write H0 to be equal to 3 into sine 30 degrees.
- Then we take all the algebraic sum of all the forces acting along the vertical.
- Then we have 1.4 is also acting upwards.
- Then the component of this, since it is acting downwards, on the other side of the equation it becomes 3 into cos 30 degrees.
- Plus we should also take the sulfate of the beam, which is 500 into 9.81.
- Okay, divided by thousand because I have to convert it into kilonewtons.
- With the values by using the equations of equivalent.
- Hope all of you have understood this.
- The next problem that I have chosen is 3 bar 30.
- A former student of mechanics wishes to weigh himself, but as.
- Only to a scale A with capacity limited to four hundred newtons and a small eighteen newton spring dynamometer B with the rig shown.

Example or application from the lecture

- The statement of the problem is given here, which goes something like this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

Core explanation · 10:00–14:59

He discovers that when he exerts a pull on the rope, so that B registers 76 Newtons.

Key details

- The scale A reads 268 Newtons.
- What are his correct weight W and mass M?
- We cannot measure anything more than 400.
- And also the maximum capacity of the spring dynamometer is 80 newton, so we cannot measure anything more than 80 newton.

- But when he is doing this experiment, he finds that the scale will read two sixty eight.
- The maximum is four hundred, the scale is reading two sixty eight, it is still possible.
- It is not more than four hundred.
- The rope dynamometer will indicate seventy six newtons, which is also less than eighty, so that is also possible.
- The force FP is seventy six newtons.
- The F of force that is the scale rating is two sixty-eight.
- What we are supposed to find is the weight of the person and the mass of the person.
- The tension that is applying here is 76 newton.
- The weight of the person, which will be W, that is acting at the center of mass of the person.
- W is equal to M into G, then the scale reading which is at the bottom of S shoes, that is 268.
- Then this tension, if you observe, there is tension here which goes here, that this would also be D.
- The same T is continuing here, so there will be T plus T, the same T is continuing here, and thus this is same.
- This would be equal to 40.
- So if I take some of all the forces, F is equal to 0 because there is only vertical forces that is acting.
- 3 bar 15, the statement of the problem is what mass mb will cause the system to be in equilibrium.
- Neglect all friction and state any other assumptions.
- There could be some small fraction that's going to be happening.
- I can resolve this along.
- the plane that is this one and perpendicular to the plane.

Example or application from the lecture

- And since he is holding this, a part of his weight will go here, which is seventy six, that is also given, that is attention.
- And that is what is given written here as 4 into 76 newtons.
- Okay, let's move on to the next problem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 15:00–19:59

The component of the sulphate acting parallel to the surface will be 15 to 9.81 into.

Key details

- Then this would also come.
- That is what is the assumption that I am making.
- MA, T da, what is to be find?
- M B solution?
- Assumption all cables attach.

- 2 L B or vertical as I told you, I am assuming that to be vertical.
- That is this tangential axis.
- This is normal axis, T axis is equal to 0, so 2t.
- is equal to 15 to 9.81 sine of this angle 25 I'll get the force to of this the component of the sulphate.
- We will go to the next problem which is three bar nineteen.
- The statements goes like this.
- When the point zero five kg body is in position shown, the linear spring is stretched 10 mm.
- Determine the force P required to break contact at C.
- Complete solutions from A, including the effect of the weight, B neglecting the weight.
- The centroid will be passing through this because it is symmetrical.
- And somewhere along this line, some point it will be acting.
- Next he says the linear spring is structured by 10 mm.
- So if I multiply 10 divided by because it is in M I have to convert it into meters so divided by thousand so we get it as 1.7 into 10 which will be 17.5 into 10 would be the force that is going to be acting which will be pulling B to the left.
- So for this condition, we have to determine what would be this force B.
- Then the self rate, which will be 0.05 into 9.81.
- Into the distance is 0.06, which is in meters, so we get it in Newton meters.
- Then the force here in this spring, which will be 70.5, into this distance is 0.04, which will create a clockwise movement, so that is why it is negative.

Example or application from the lecture

- And that is 0.05 is the mass that is given, so I multiply with 9.81 and.
- That would be the weight of the cell fate of the body that is given, which is 0.05 to 9.81, which will be passing through this line.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

Core explanation · 20:00–24:59

Then neglecting the weight, that means we have to remove this.

Key details

- Hope everybody has understood this.
- The next one is 3 bar 20.
- So there is a clockwise movement here which is going to be acting because of the spring, which will be equal to 0.7 newton meters.
- We are again supposed to find out what is the value of P when there is no contact here.
- The self weight also we know, but there's a clockwise movement here, so minus equals 0, so we get P.
- Neglecting the weight, we have to neglect this.
- And take it so we get P as 6.25 newtons.
- We will be continuing in the next class.

- We will solve some more problems in the coming classes.
- My name is Dr.
- Sviving Tesh from PS University.
- We'll see you in the next.

Example or application from the lecture

- This is similar to the previous problem except that instead of a spring that was attached here, linear spring, now they have given you the torsional spring attached at O.
- And they have given that the torsional spring at goes pre-tention so that the so that it exerts a 0.75 N meter clockwise moment.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

5 UE26ME142A_Unit 2_Class 17_AVSummary

Study focus: UE26ME142A_Unit 2_Class 17_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000058 · EV0000059

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Svink Tesh from Civil Engineering under the Department of Science and Humanities PS University.
- This is class number 21.
- problems on equations of equilibrium.
- Which states that the ATKG exerciser is beginning to execute some slow, steady bicep currents.
- As tension T is equal to 65 newton is developed against an exercise machine not shown.
- Determine the normal reaction forces.
- feet A and B friction is sufficient to prevent slipping and the exerciser maintains the position shown with the center of mass at G.
- Under the feet A and B, and he also says that the friction is sufficient to prevent slippage.
- There's a tension T, there's a machine here which is not indicated.
- The direction of this tension T with respect to the horizontal is 20 degrees.
- The centre of mass of the person.
- is G where the 80 kg mass is going to be acting and of course if you convert it into a weight it will be 80 into 9.81 newtons which will be acting at G.
- Then you have two points are the under the feet of the person.

- Point A and point B, you will have a normal reaction exerted at A, the normal reaction exerted at B and there will be a friction force.
- One is the tensile force acting to the right of the screen, which is 65 newtons.
- Then the sulphate of the person which is 80 kgs into 9.81, which will be acting at point G.
- You will have a normal reaction N acting upwards at A, normal reaction at B acting upwards, frictional force at A which is acting to the left of the screen, frictional force at B acting to the left of the screen.
- All the forces that I have discussed is indicated here along with the dimensions.
- Then the sign convention is also indicated here.
- X is positive, y is positive up third, x is positive to the right.
- Then movement, anti-clockwise movement is taken as positive.
- The horizontal component of the tension will be equal to the frictional force that is going to be acting F A and F B, but we are not interested in that.
- What we are interested is in N A and N B.
- There are two unknowns.
- The horizontal summation of all the forces along horizontal cannot be taken, so we can take summation of all the forces along vertical algebraic sum.
- Then we can take movement about any point to be equal to zero.
- So first equation summation of all the forces acting along Y to be equal to zero.
- Plus or this is negative, NEA is positive, N B is also positive, this is negative, then the component of this T will be acting upwards, so it is positive.
- That is 65 sin 20.
- That should be equal to 0 and that is what is indicated here.

Example or application from the lecture

- look at what is given in the statement.
- They have given you the mass of the person, which is 80 kgs.
- They have also given you what is the tension in the cable, which is T that is equal to 65 newtons, and what is asked is to determine what is the normal reaction.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

Core explanation · 05:00–09:59

Next taking moment about any point, it is better to take the moment at the point where you have the support reactions.

Key details

- We get NA.
- into the distance between A and B which is 500 millimeters.
- Then you have this sulfate of the person is 18 to 9.81 into the horizontal distance between G and B is 250.
- Then we have.

- The component of T along the horizontal, which is 65 into cos 20 into the vertical distance, that's what is indicated here.
- Then we have a component of T which is acting along the vertical, which is 65 sin 20, into the distance from B to C, which is 350 mm, and that's what is indicated here.
- We will get NA equals 219 from this.
- Substituting the value of NA here, we get the value of N B which is equal to 544 m.
- I hope you have understood here.
- Of course the algebraic sum of all the forces along the horizontal if you take, you will have FA plus F B to be equal to 65 cos 20 degrees.
- I hope all of you have understood this.
- The statement of the problem is determine the moment which the motor e must exert.
- In order to position the uniform slender bar of mass m and length L in the arbitrary position theta, the ratio of the radius of the gear whe attached to the bar.
- To that of the gear wheel A, attach it to the motor shaft is 2.
- What am I supposed to apply the movement here such that this bar of length L would be in the position in an arbitrary position.
- That is at an angle theta is what is asked.
- If this has to rotate in the anti-clockwise direction, then this arm will also rotate along with that and it will come to the or in this arbitrary position at an angle T.
- If this has to rotate in this way,.
- Then the movement that needs to be applied here, so here will be in the clockwise direction.
- When I apply a movement here, so when I apply this movement, you will have a force that is going to be acting along this tangent line.
- The weight of this would be acting at the center.

Example or application from the lecture

- No, if there is no movement say consider at this point, then this arm would simply rotate in the clockwise direction and this will come to a vertical position.
- No, they have also given you the ratio of the radius of the shaft that is this and this or in the ratio of 2, or rather if I take this as R, then this would be 2 R.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

Core explanation · 10:00–14:59

The sulfate of the of this the O C would be acting vertically down which is equal to Mg.

Key details

- The force that is acting on the gear would be like this, which is F, because this would produce a clockwise movement to prevent it from rotation, I should apply a force F here, which will act in the anti-clockwise direction.
- sigma m or b is equal to 0, f into 2 r minus mg into this horizontal distance would be this is L by 2 sine of theta that's what is indicated here.
- M, G, and L as well as T.

- Now this force F which is acting here would be generated because of the gear system at A.
- Here, the fourth that is going to be acting here will be in the opposite direction.
- Whereas this force would act in the anti-clockwise direction.
- We will get this as R and R will get cancelled.
- That should be the moment that should be produced in the motor that is present at the point A, that is this one.
- You have the motor, the movement that should be produced in this motor, which should rotate in the clockwise direction, should be equal to one fourth of your MGL sin theta.
- Then it is also depends upon at what angle this bar is to be rotated.
- I hope you have understood.
- The equations of equilibrium that is summation of the forces along X, algebraic sum of all the forces along X, algebraic sum of all the forces along Y, then movement about any point should be equal to 0.
- But before using these three equations,.
- I am able to draw a clear free body diagram, right?
- Isolating what is that I want.
- As you can see here, I have taken the pulley at B and the rod that is acting at B separately along with the forces that is acting on this body.
- That is the force of.
- The sulfate of the rod okay is what I've taken.
- What are this B will now act as an hinge.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

Core explanation · 15:00–19:59

Okay, so this also you can see that it is like an hinge, and because of that, you have these forces a y and ax.

Key details

- Of course, nowhere I have taken.
- By, Bx as well as AY and AX, because in both the cases what I have considered is summation of all the movement about B is equal to 0, summation of all the movements about A is equal to 0.
- I have not used the vertical force, algebraic sum of vertical force, or algebraic sum of the horizontal forces because that is not what I intend to determine.
- They have not asked us to determine Bx or BY as well as Ax or AY.
- If they would have asked, I can easily use summation of all the forces along vertical to be 0, summation of all the forces along horizontal to be equal to 0, and then I could have determined what is Bx and BY.
- Since I have got F here, okay, this F.
- And I can also determine what is the angle made by this force with horizontal or vertical.
- I can resolve this along vertical and horizontal, and I will get what is Bx in terms of F and also BY in terms of F and the sulfate of the rod.

- But in the problem, since they have not asked us to determine the reactions acting at B or A, we are not finding out that.
- A statics by Miriam and Craig and Bolton published by Wiley Publication and it is 8th edition.
- Indian edition itself.
- We'll be continuing solving of few more problems in the next class.
- My name is Dr.
- Sri Vengtesh from Department of Civil Engineering under Department of Science and Humanities, PES University.
- We will meet in the next class.

Example or application from the lecture

- Okay, and all the problems that are solved are example problems from the prescribed textbook, that is engineering mechanics.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

6 UE26ME142A_Unit 2_Class 18_AVSummary

Study focus: UE26ME142A_Unit 2_Class 18_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000074 · EV0000075

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Isviving Tish from Civil Engineering under the Department of Science and Humanities, PS University.
- In the last class, that is class number twenty one, we had discussed two problems on equilibrium.
- Today we will be discussing some more problems.
- So first one that I have taken as 3 bar 27.
- The statement of the problem is find the angle of tilt theta with the horizontal so that the contact force at B will be one half that at A for the smooth cylinder.
- The statement what we could identify as that we need to find out the angle theta for the condition that the contact force acting at B is equal to one half of that acting at A.
- On this V block, they have kept a.
- If the mass of the cylinder is M, then the sulphate of the cylinder would be mass times the accuration due to gravity, which will be acting at the center.

- The contact point of the cylinder on the V groove is at two point, that is A, that is this point, and B that is this point.
- The self weight will be acting downwards.
- That has to be balanced by the reaction.
- You can know that this line is the tangent to the cylinder.
- Because that would be only at one contact point, and that would be always going to be tangent to the cylinder.
- So the reaction, if it is normal to this, then that means it would be passing through the radial line.
- Which will also pass through the center of the cylinder.
- This reaction at B is going to be normal to this line or this groove, which is also tangential to the cylinder.
- Which will also pass through the center of the cylinder.
- So you will have three different forces acting because there are only two contact points where the cylinder is touching the V group, that is at A and B.
- Sulfate of the cylinder that is going to be acting.
- Next, to know the angle at which the reaction at A and B is acting, we have to identify this, which is 45 degrees.
- The angle made by the group, V Group, which is passing through A.
- With respect to the horizontal, we'll be at an angle of 45 degrees minus this angle theta, which is 45 minus theta.
- Whereas here, if I you find out the angle made by the group passing through B with the horizontal, that would be equal to 45 plus theta.
- If I draw a.

Example or application from the lecture

- That is this one, this line, is also given and that angle is forty-five degrees, which you have to note down.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 05:00–09:59

The reaction, if I draw the reaction here, this total angle will be 90 degrees.

Key details

- Since this would be 45 minus theta, the angle made by the reaction at A with the horizontal will be 45 plus theta.
- And this angle is 45 plus theta, and this reaction will be at 90 degrees to this line.
- The angle made by the reaction with the horizontal will be 45 minus theta.
- We have removed the V group.
- The reactions at A, the reactions at B is indicated.
- The X and Y axis are indicated.
- You can see the sulphate of the cylinder is indicated as Mg, that is reacting vertically down.

- Next with the horizontal is going to be 45 plus theta.
- I have explained how to get that.
- I have explained that also.
- I hope it is clear till now.
- You have to identify that the reaction at A is considered as A itself, the reaction at B is considered as B.
- Which is equal to one half of that A.
- Reaction at B has to be one half of A.
- For that condition, we have to find out what is this T.
- At the center of the cylinder.
- By considering the reaction at B, this is reaction at B which is equal to one half of K.
- This is the reaction at A, which is A itself.
- This is the cell fate of the cylinder that is Mg.
- would be 45 plus theta is this minus 90 minus 45 plus theta so this would be 45 minus theta that is what is indicated here.
- Tan of 45 minus theta, which will be equal to the opposite side by adjacent side, which is B by A, and we know that B is one half of A, so it is A by 2 divided by A.
- But here instead of taking the distances, we have taken the reactions as the size.
- Or the force at the sides of the triangle, which will still work.
- You can represent the distances by forces or force by distances.
- However, if you want to solve it by using the equation of equilibrium, you can also solve it by taking.
- Sum of all the forces along X to be equal to 0.
- That means your force F cause of 45 minus Teta, which would be this, would should be equal to the reaction V cause of this.
- Cause of forty-five minus theta which is this angle.
- The value of theta.
- Along with that, we can also use some of all the forces along acting along y.

Example or application from the lecture

- Let's continue.
- That is what is given in the statement of the problem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 10:00–14:59

B sin 45 minus theta.

Key details

- You can use that also to solve this.
- Hope you have understood.
- I've considered problem 3 bar 31.

- The statement of the problem states determine the minimum cylinder mass m1 required to cause loss of contact at B.
- This mass m1 is attached to this cylinder.
- The cylinder is.
- Hanging by this row or wire.
- The wire is attached to this beam.
- One end of the wire is attached to this point beam.
- You can also see that this is an R.
- The mass of this entire.
- AOL the L angle or L member is to be considered as M.
- Whereas the mass of OB will be equal to M by 3.
- The force will always be acting vertical.
- That is 2 by 3 m times g will be acting at the center of AO, which would be at a distance of 2 by 3 L divided by 2, that would be L by 3, whereas the mass m by 3.
- Would M G by three will be acting at this point center and this distance will be equal to L by three into half or into half, which will be L by six, that distance where the force is going to be acting.
- Then you can see that here.
- The point O, there is an hinge, so there will be two reactions, one vertical and one horizontal.
- Make the contact here.
- There should not be any contact.
- So that means the reaction at A should be equal to 0.
- The free body diagram for this M1, if you take if I take the tension in this row as T.
- The same rope is continuing here, so you will have T here also.
- M1 by M1 G by 2, that is what is the force that's going to be acting here.
- This force will be acting downwards, which will make this two rotate in such a way that there is no contact here.
- So if I take some of all the vertical forces acting on the mass m1g which I have explained it just now, so I get t to be equal to m1g by 2 because you have 2t here just to be balanced by this m1g.

Example or application from the lecture

- Let's go to the next problem.
- So the condition that is given is there should not be any contact at A.
- They have given the inclination of this arm AO with respect to vertical as 30 degrees.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

Core explanation · 15:00–19:59

Then you have the sulfate of this R which is 2 mg by 3.

Key details

- Then the sulphate of OB which is mg by 3 because this length and this length are 2 L by 3 and L by 3.
- Then we also know that this length the inclined length of OA is L by 6.
- This distance is L by 6.
- This inclination is 30 degrees, so obviously this will be 60 degrees.
- This if you observe AOB is at an angle of 90 degrees.
- We know that this is 30, so if I draw a vertical line here, this would be 30, this would be 60, so this obviously has to be 30 degrees.
- Because we know that there are two reaction that is acting here, which are unknown.
- So I have to use the first equation of the equilibrium, which is the moment equation, at the point where you have the maximum or not.
- Sine 30 degrees, and that's what is mentioned here, which will produce an anti-clockwise moment, so positive.
- This would produce a clockwise moment, so negative.
- This force is mg by 3 into the horizontal distance of this force from O will be L by 3.
- Cost 30 degrees because this angle is 30 degrees.
- Then we also have this tension T which will also produce a clockwise movement.
- The horizontal distance of B from O.
- will be equal to L by 3 into cos thirty degrees and that's what is mentioned here.
- So if I simplify this I will get the value of M1 because M will in terms of M, M1 we will get it in terms of M.
- Hope everybody has understood this.
- That's why we are using only the moment equation here and we are stopping it.
- But the extension of this problem can also be if they ask you to determine the reactions are to O.
- Then if I use summation of all the forces along y to be equal to 0, so I'll get the value of OY which will be equal to m into g, summation of this, plus this T would be the reaction that's going to be acting at OY.
- Now O X is 0 because there is no.
- Horizontal force that's going to be acting here because this was anyhow zero.
- Hope everybody has understood this two problems.
- Whatever I have discussed.
- What we need to do is only use the equations of equilibrium, the three equations of equilibrium, algebraic sum of all the moments to be equal to zero at any point.
- Then summation of all the forces along y should be equal to zero.
- The algebraic sum of all the forces acting along y direction to be equal to zero, the algebraic sum of all the forces acting along x direction to be equal to zero.
- And that's what we have done now.
- But you can use only the earth equation, whatever is required for you to determine the unknown value, okay?
- So we'll stop today's class here, we'll continue in the next class.
- My name is Dr.
- See you in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

7 UE26ME142A_Unit 2_Class 19.0_AVSummary

Study focus: UE26ME142A_Unit 2_Class 19.0_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000090 · EV0000091

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

This is class number thirty twenty-three, I'm sorry.

Key details

- This is class number twenty-three and I'm Dr.
- Sv Venkitesh, Civil Engineering.
- Department of Science and Humanities under PS University.
- In the last class we had discussed about the problem three bar 31, I think.
- I have considered this problem, which is three bar thirty-two.
- The statement of the problem is cable A B passes over the small ideal pulley C without a change in its tension.
- What length of cable C D is required for static equilibrium in the position shown.
- What is the tension T in cable C D?
- Which passes over a small pulley C without changing its tension.
- That means the tension in A C and the tension in C B would remain the same.
- That's what it means.
- Your ACB, this is the cable.
- The mass 20 kg into acceleration due to gravity.
- For this condition, that means when this angle is 45, what would be the length of C D is what is asked.
- For this condition,.
- Mean you need to determine what should be the length of C D and also what is the tension in C D.
- And C B.
- So since this is forty-five degrees, this would also be forty-five degrees because that is alternate angles.
- The tension in this cable would be the same as the tension in cable C B, which is 20 into 9.81, that is how it is include indicated here.
- The tension in C D is indicated as T, the angle made by C D with respect to horizontal is considered as theta.
- Then the tension in this cable is again twenty to nine point eight one.

- And this distance is six hundred and now with this if you observe this there are three tension acting at a point.
- Then what is the sine rule or Lamis theorem?
- Any force that you take where you have three forces acting at a point.
- The force divided by the sign of the angle between the other two forces is a constant, or it is equal to the other force divided by the sign of the angle between the other two forces.
- We know the tension in this.
- We know this is forty-five, this angle would be 90 degrees.
- This would be equal to 90 plus theta, whereas this would be equal to 180 minus 45 minus theta.
- This force divided by this angle, or this force divided by this angle.

Example or application from the lecture

- Input, he has asked you to determine what should be the length of the cable CD, okay for static equilibrium for the given position, what is the tension T in the cable C D.
- Then he is also given in the diagram that this angle has to be 45.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

Core explanation · 05:00–09:59

The distance C D once I get this angle theta, I know this distance.

Key details

- Then 20 into 9.81, cos 45 will also be along X, but it is 9 minus.
- From this equation, I can easily find out what is T.
- Then you take all the sum of all the forces along y direction.
- cost forty five.
- Or cos 45 or you can also take it as sin 45 because since it is 45, this would also be 45.
- So 20 into 9.81 sin 45 is positive because it is acting upwards.
- Then t sine theta, then that would be the component along vertical, that would also be positive.
- Then this one is negative, so that is twenty into nine point eight one equals zero.
- Since we have got T value from this equation, I can substitute here and find out what is angle theta.
- Using this angle theta and this triangle, right angle triangle, I can take tan theta to be equal to.
- or rather I should take it as sine theta which is equal to 600 by Cd.
- Using these two equations the two unknowns which is T and Teta has been determined.
- Once you get the.
- Value of theta you can use the triangle C D C D and this triangle.
- Hope all of you have understood this.
- I have considered problem 3 bar 34 here.
- The statement of the problem goes like this.

- The small slider A is moved along the circular slot by a mechanism.
- Attached to the back side of the rectangle or plate for the slider position theta equals 20 degrees, shown determine the normal forces exerted at the small stops C and D.
- The unstretch length of the spring of constant k, which is equal to 1.6 kN per meter, is R by 3.
- The value of R is 25 mm, and the plate lies in the horizontal plane.
- neglect all friction.
- The spring constant K is 1.6 kN meters.
- The unstretched length of the spring is R by 3 and R is equal to 25 mm.
- The radius R is this.
- The unstretched length of the spring is R by 2.
- That means if I don't stretch this, okay, if I don't elongate it, the length of the spring will be R by 3.
- Okay, because it is A B the stretched length.

Example or application from the lecture

- You can solve it either using sign rule or Lamy's theorem, or you can resolve the given forces along vertical and horizontal, and then resolve.
- Here the solution, they have resolved the given forces along horizontal and vertical, then they have equated all the forces along vertical, and all the forces along horizontal, and then.
- So let's move on to the next problem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000094 · EV0000095 · EV0000096 · +1 source units in pack.json

Core explanation · 10:00–14:59

The two stoppers is given here, which is C D.

Key details

- Because of this, a component of the force will be acting upwards, a component of the force will be acting to the right.
- Because of this component at this position, which is an hinge here, so this would try to rotate this plate in the counterclockwise direction.
- That would be prevented by the stopper here, so you will have a reaction at D.
- Whereas here there won't be any contact, so you will not have any reaction.
- But if this spring is moved above this horizontal in any position here, then you will have the.
- Tension in the spring acting like this from A to B.
- So because of which you will have a component acting downwards which will make the plate rectangular plate to rotate in the clockwise direction.
- At the top point you get the contact at C.
- I get an horizontal reaction here, a vertical reaction here, the sense can be anything.
- Then the force acting at A, okay, which should be at an angle of theta with respect to the horizontal that is this angle theta so you will have two components here one will be whatever

is the spring force f s this would be sine theta this would be cos theta then we will have a reaction at D which will be acting downwards okay.
- I should determine the.
- Normal force exerted at C and D, that means at D, what is the normal reaction exerted?
- So this won't come into picture because we are taking the forces acting on a plate only.
- There's a spring force F S okay, which will be equal to K times delta.
- What is Delta R is equal to 25 mm which is 0.025 meters, F S is K delta.
- K into what is delta as I have explained it, it is R minus R by 3.
- By inspection here the reaction is 0, you get the reaction at D only.
- Horizontal distance, we need to calculate that.
- How do we calculate that?
- Not this testance, we can calculate it as this is R.
- If I take the component of this along the horizontal, so I get this as R cos 20, which will be this.
- 3 by 2 R because 1, 2, 3, 3 R by 2.
- Would be this total R, this total is.
- R by 2, so that you have to take into account.
- I hope that is clear.
- What is clear?
- This total distance is 3 by 2 R.
- That is 3 by 2 into R is 0.025.
- Then minus R then we have this force D into this distance which is R which is 0.025.
- This is 1.5 R.

Example or application from the lecture

- The component of this force acting along the vertical is 26.7 sin 20 because this angle is given as 20 degrees.
- Then they have given this distance as R by 2 and this also as R by 2.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000098 · EV0000099 · EV0000100 · +1 source units in pack.json

Core explanation · 15:00–19:59

That should be equal to 0.

Key details

- I hope all of you have understood this.
- If there is a doubt, that would be equal to total distance, which is 3 by 2 r minus this distance, which is r cos 20 degrees.
- 26.7 sine 20 is this into this distance which will be this component.
- There's one more component here.
- This is F S cos 20 degrees into this distance which will be equal to R.
- Cost 20 degrees.
- I hope that is clear to all of you.

- You can write down that and check it.
- Cause of that, okay, we will get this distance the sign of that we will get this distance.
- You can do it in the other way, okay.
- If I directly want to take this force into this particular distance, that's one way of solving it, or I can resolve this force, okay, or I can take it to be acting at B itself.
- Sine 20, which will be acting here, the total distance from here to here will be 3 by 2 r, and that's what is taken here.
- Is Fs sin 20, which will be acting here.
- The distance from here to here will be 3 by 2 R.
- Then they have taken this force into this distance.
- Solving this problem by considering the force FS to be acting at B, or you can take the force to be acting at A, resolving that along vertical and horizontal, or you can take this force and find out what is the perpendicular distance.
- I have talked about all the three different ways of solving it.
- It is clear.
- I will explain it to you.
- We will continue this class in the next class.
- That is class number.
- And I am Dr.
- Sv Vintesh from Civil Engineering under the Department of Science and Humanities under PS University.
- See you in the next class.

Example or application from the lecture

- But what is given here, the equation, is you are taking the force Fs to be acting at B, only this component will come into picture, whereas the horizontal component, since it is passing through O itself, it doesn't have the perpendicular distance, so that would not come into picture.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000102 · EV0000103 · EV0000104 · +1 source units in pack.json

8 UE26ME142A_Unit 2_Class 19.1_AVSummary

Study focus: UE26ME142A_Unit 2_Class 19.1_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000106 · EV0000107

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

This is class number twenty four and I am Dr.

Key details

- Esvin Venkatesh from Civil Engineering Department under Department of Science and Humanities, PS University.
- In the last class that is class number twenty-three, we have.
- Worked out two problems on equilibrium.
- We'll continue to solve some more problems in today's class.
- That is class number twenty four.
- I've taken a problem three bar thirty-five, the statement of the problem goes like this.
- The asymmetric simple truss is loaded as shown.
- Determine the reactions at A and B, neglect the weight of the structure compared with the applied loads is knowledge of the size of the structure necessary.
- Truss is a structural member which is used for carrying out load from the roof to the columns that is supporting this truss.
- We are supposed to determine the reactions at A and D.
- But to neglect the weight, sulfate of the structure itself, because that is going to be very, very small when compared to the load that is coming on the process.
- And they have also asked you to.
- Let them know whether the size of the structure matters or not.
- And there is a load of L acting vertically down.
- At C there is a horizontal load which is acting to the left, which is negative of the.
- Then there's a load at D which is acting vertically down that is also negative.
- Then you need to identify that the support at D is an hinge support.
- This angle that is the angle made by the side B with EA is also sixty degrees.
- That makes this as an equilateral triangle.
- There were also even this angle as 60 degrees.
- Does the size of the structure matter or not?
- Roller support at A by a reaction which is represented here as AY.
- Replace the reaction at D which is an in support by the reaction.
- There will be two reactions, one vertical and one horizontal.
- Then apply the same forces whatever is there acting at different point at this.
- The sign convention, x axis positive to the right, y axis upwards positive, and anti-powered rotation as positive.
- So when I represent that as D, obviously your AB and BE will also be having the same distance because that forms an equilateral triangle.
- Next, if you observe ABD, this triangle.
- C D E.
- This is sixty degrees, this is thirty degrees, so this has to be ninety degrees.
- They are similar triangles.

Example or application from the lecture

- And if you read the diagram that is given, this is the diagram that is given.
- Axis that is given here, the positive axis X and Y.

- Then they have also given you the angle made by AB with respect to the horizontal which is EAE that is sixty degrees.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

Core explanation · 05:00–09:59

If this is D and this triangle C D E and ABD are similar.

Key details

- So if this is ninety degrees, this is sixty, sixty, I can get this angle as sixty itself, because the total would be one eighty.
- What is remaining is?
- 180 minus 90 minus 60 is 30.
- Cos 60 degrees, which will be D by 2, that is what is mentioned here.
- Next, using these similar triangles, since they are similar triangles, if this is D, this E point.
- Okay, as to be the midpoint of this triangle because A B should be proportional to BE.
- AB should be proportional to E C.
- And that's what is mentioned here.
- One is the moment about any point is zero.
- Then you can take sigma f.
- X to be equal to 0, that is sum of all the horizontal forces should be equal to 0.
- Then if I take some of all the vertical forces to be equal to 0, I get.
- A y and dy to be equal to L plus L plus L 3L.
- Or if I take the moment about D, I would have I would get the value of AY.
- So first sigma f x equals zero sum of all the forces.
- Which is in the negative direction, so dx minus l equals 0.
- Then you take sigma f y equals 0.
- To be equal to L plus L plus L as I have told you, so that is acting in the downward direction.
- Then taking moment either I can take D as the moment or A as the moment because here already we have got the value of DS.
- That would be anti clockwise or positive plus L.
- That is this L which will produce an anti-clockwise movement at A.
- This would be the perpendicular distance.
- This angle is 60, so sine 60 I get this distance.
- That is root 3 by 2.
- into D by 2 this L would produce a clockwise movement okay but this L into the perpendicular distance would be this which is definitely D by 2 because this is an equilateral triangle so this would be the midpoint of this base A.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000110 · EV0000111 · EV0000112 · +1 source units in pack.json

Core explanation · 10:00–14:59

Then we have this force also, that is that would also produce a clockwise movement, so minus L into this is D, this is D, so 2D should be equal to 0.

Key details

- We will get the value of AY and DY.
- Simplifying this, I would also get the value of AY and DY in terms of L.
- Because L is the load that is applied on the structure.
- D can have any value.
- This reactions would not change, it would remain same.
- The knowledge of the size of the structure is not influencing the support reactions.
- Hope you have understood this.
- The next problem is 3 bar 37.
- The statement of the problem goes like this.
- The indicated location of the center of mass of the 1600 kg pickup truck is for the unladen condition.
- If a load whose center of mass is X, which is equal to 400 mm,.
- Behind the rear axis is added to the truck.
- Determine the lower mass M L for which the normal forces under the front and rear wheels are equal.
- The centre of mass of 1600 pickup truck.
- 1600 is the total weight of the truck.
- And the centre of mass is for the unladen condition, means there is no load there.
- Center of mass, then when there is a load that has been applied such that the center of mass of this load is at a distance of 400 mm from the rear axle of the truck, you need to determine.
- What would be the mass M L for which the normal reaction under the front and the rail wheel are equal.
- The G is the center of mass of the truck where your thousand six hundred kg low or mass will be acting.
- Then you have the mass of the load ml which is act at a distance of 400 mm from B.
- There are two wheels if you are in contact with the surface, so you will have a reaction at this point, A and B.
- What is the distance between central mass of the truck and the point B, which is 1675 mm?
- We are supposed to find out mm for the condition.
- The reaction at A and reaction at B are equal.
- There are two reactions at A, one under this wheel what we are looking at.
- There is one more wheel on the earth.
- Back side of A, so we can take it as 2 R A.
- 2 R A should be equal to 2 R B, R A should be equal to R B.
- And here he has taken that as NA, here he has taken it as N B which is equal to N.
- Both of them are equal.
- For that condition we have to find out ML.

Example or application from the lecture

- Let's go to the next problem.
- What is given in addition to that is what is the distance from A to the centre of mass of the truck, which is 1125.
- X distance is given as 400 mm.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000114 · EV0000115 · EV0000116 · +1 source units in pack.json

Core explanation · 15:00–19:59

Then you have the weight ml or rather WL that is acting on the truck, but we have to find out what is the mass ml.

Key details

- Then you have the reaction at A, reaction at B, both of them are N and N itself.
- Sigma Fx equals 0, that will not come into picture because there is no horizontal force, so no horizontal reaction.
- Then you take sigma f y equals 0, so you get 2n equals 1609.81 plus WL.
- Then you can also take moment about any of the supports, say I will take it at A or at B.
- This point I will get this N into this distance, which is thousand one hundred and twenty-five plus thousand six hundred and seventy-five, which will be in the anti-clockwise direction.
- Whereas this force will produce a clockwise movement, this force into this distance.
- Then this force into the total distance would also produce a proper moment.
- Using these two equations I can find out these two or not, which is N and WL.
- So first taking moment about point A.
- This is in the clockwise direction, so taken as positive.
- Then n which is this into this distance, you will get it as 2800 convert it into meters which will be 2.8.
- This also converting it into meter it is going to be 1.125.
- Then we have this force which will produce a clockwise moment.
- Solving this, you can see that there are two unknowns, N and W L.
- Then taking all the forces along the vertical to be equal to zero.
- You get 2n because these two are the vertical forces acting upwards should be equal to this which is acting downwards, thousand six hundred into nine point eight one.
- Then we have this also which is acting downwards, so minus W L.
- By solving these two equations,.
- I will get the value of N and I will get the value of WL.
- Because of the loaded or the load on the truck, I would divide it by acceleration due to the graphic to get the mass of the load that is added.
- such that the reactions at A and B are equal, which is 44 kgs, and the center of this mass should be at a distance of 400 mm from B.
- Hope you have understood this problem also.
- This is class number twenty.
- Or we will continue the same thing in our next class.

- We'll try to solve few more problems on equilibrium chapter and try to understand how we solve problems.
- using the equations of equilibrium.
- I am Dr.
- SV Vengtish from Civil Engineering Department of Science and Humanities PS University.
- See you in the next class.

Example or application from the lecture

- Same thing is what is given here.
- That is also given in the problem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000118 · EV0000119 · EV0000120 · +1 source units in pack.json

9 UE26ME142A_Unit 2_Class 19.2_AVSummary

Study focus: UE26ME142A_Unit 2_Class 19.2_AVSummary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000122 · EV0000123

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Svingtish from Civil Engineering under the Department of Science and Humanities, PS University.
- This is class number 25.
- We are solving problems related to equilibrium.
- We will continue to solve some more problems on this chapter equilibrium.
- The statement of the problem goes like this: a block placed under the head of the claw hammer as shown greatly facilitates the extraction of the nail.
- If a 200 newton pull on the handle is required to pull the nail.
- Calculate the tension T in the nail and the magnitude A of the force exerted by the hand or head on the block.
- The contacting surfaces at A are sufficiently rough to prevent slippage.
- Go through this statement, what is that we can understand as there is an hammer used to pull out a nail.
- A force of 200 newton is applied onto the handle.
- They have asked us to determine.
- The tension T in the nail and the magnitude A of the force exerted by the hamorate on the block.

- That the force applied onto the hammer, that is 200 newtons, is perpendicular to the axis of the hammer, which passes through A, and this axis of the hammer is at an angle of 20 degrees to the vertical.
- The distance between point A and the point where the force is applied is 200 mm along the axis of the hammer.
- The nail is at a distance of 50 mm along the horizontal from point A, and at A,.
- The force applied on the hammer, 200 newtons.
- The distance between the force and the point A.
- Then the tension in the nail.
- The tension means the nail has been pulled.
- Then at point A we will have a normal reaction which is passed through the axis of the hammer and a friction which will prevent the.
- Slipping of the hammer.
- The 200 newton force acting on the angle of the hammer.
- the angle made by the axis of the hammer with respect to the vertical 200 twenty sorry twenty degrees.
- Then the distance between the force 200 newtons and point A, that is 200 mm, the tension in the nail which is acting downwards because it's the internal force P and the distance between A and this tension should be.
- As I told either you can take the.
- Normal reaction acting on the hammer along the axis of the shaft and the frictional force.
- Or we can also take the vertical reaction and the horizontal reaction that is acting at A, considering it to be a rough surface.
- Then take all the forces acting along the x direction and equally to 0.
- Then we have a force of 200.
- Cost 20 degrees because this angle would be 20 degrees with respect to the horizontal.
- 200 cos 20 degrees.
- Only two forces are acting along the x axis.
- Then taking all the forces along y, we have a by acting upwards minus t.

Example or application from the lecture

- What ES given is that it is the friction is quite sufficient the or the roughness is sufficient to take up the contacting surfaces at A.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000122 · EV0000123 · EV0000124 · +1 source units in pack.json

Core explanation · 05:00–09:59

Then a component of this which is also acting upwards, which is two hundred sine twenty degrees, that's what is given here.

Key details

- substituting the value of T as 800 newtons, we get AY which is equal to 725 newtons.

- Then we can find out what is the resultant of Ax and AY, which will be A which is equal to 755 newtons, which will be acting one force reaction at A.
- Hope everybody has understood this problem.
- There are three equations and what we are finding out is also three unknowns.
- The three equations of equilibrium is algebraic sum of all the movements about any point should be equal to zero if the body is in equilibrium.
- Then algebraic sum of all the forces along X direction or the component of all the forces along X direction should be equal to zero.
- The algebraic sum of the components of all the forces acting along y direction should be equal to zero.
- And that's what we have used to solve this problem.
- Hope everybody has understood this problem.
- 3 bar 43.
- The statement of the problem goes something like this.
- In a procedure to evaluate the strength of the triceps muscle, a person pushes down on a lower cell with the palm of his hand.
- in the figure.
- If the load cell reading is 160 newton, determine the vertical f tensile force generated by the triceps muscle.
- The mass of the lower arm is 1.5 kg with mass center at the G.
- State any assumptions.
- The force is applied by the palm of the hand.
- No, if you push your hand in air, you won't feel any force in the tricept muscles.
- But if you have something which prevents then if you try to.
- You can try this.
- If you try to push it, you will you can feel that there is there is some strain or you can feel the force in the bicep muscles.
- He has asked you to determine the vertical tensile force in the tricep muscles because of the force that is exerted by the palm onto the low side.
- Whatever is the force that you are applying by the palm, okay, there is that force is coming because of the tricep muscles.
- What would be the force experienced by the tricep muscles?
- Because of this force that you are generating.
- At the load cell.
- Then he has told to state any assumption.
- Then you have the point G which is at a distance of 150 mm from the load cell, horizontal distance.
- Then we have the joint in the arm.
- The joint is at O.
- The distance from G to that joint is 150 mm.
- This joint O will act as an hinge.
- Because you are able to move the arm.
- Because of which you have apply the force there.
- There will be two reactions, one vertical and one horizontal.

- which gives you where the tricep muscle is.
- That's the assumption.
- That's what he has told.
- State any assumption the assumption is that.

Example or application from the lecture

- Then they have given the small distance 25 mm.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000126 · EV0000127 · EV0000128 · +1 source units in pack.json

Core explanation · 10:00–14:59

We are assuming that the tricep muscle is vertical.

Key details

- Then G that is the way mass is acting downwards.
- Then you have the reaction at O, then tricep muscles which will be acting upwards.
- Very important thing is you have the joint at O which will act as an hinge.
- The mass of the arm is acting G, 1.5 kg into 9.81.
- The lower cell, the sense of the lower cell, when you apply this force, the reaction that you get it is acting upwards.
- That is why you get the pain in the tricep muscles and that is acting up force.
- That sense of that force 160 un you have to clearly mention and or take it to be acting of course.
- Then only you will be able to get the force in the tricep muscles.
- Very simple free body diagram.
- Along with the forces, the distances of each and every point, that is from O to the force in the tricep muscle or the tricep muscle itself is 25 mm, O to the G point, the horizontal distance is 150 mm.
- Then G2, the low cell point is also 150 mm.
- They have asked you to determine the force in the tricep muscles only.
- I get f into 25 which will produce a clockwise moment.
- Then we have the self rate of the R, which is 150 to 9.81, into the distance, perpendicular distance is 150, which will also produce a clockwise moment.
- This would produce an anti-clockwise moment at point O.
- What about the other two?
- Moment at point O only.
- And at point O you have the unknown as OY and O X.
- But in the problem since they have asked only the force in the tricep muscle, we are not bothered about OX and OY.
- Further, if you would have asked you to determine the reaction that you get at O, then you should have taken sigma f x equals 0, sigma f y also to be equal to 0, and then you should have calculated the value of Ox and OY.
- Since that is not asked, we are not interested in the other equations.

- You would have got F into 150 plus 25, which is 175 as the clockwise moment.
- Then you would have got OY into 150 as again the clockwise moment.
- Then you would have got 160 into 150 as the anti-clockwise moment.
- Minus 1.5 into 9.81 plus 160 to be equal to 0.
- By solving those two equations, we could have got the value of f as well as OY.
- However, the value of Ox would be 0.

Example or application from the lecture

- I will consider that as negative.
- Now other than point O, if you would have taken the moment about any other point, say for example, if I would have taken the moment about G.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000130 · EV0000131 · EV0000132 · +1 source units in pack.json

Core explanation · 15:00–19:59

Because there is no horizontal force that is acting on the arm.

Key details

- So when there is no action, there won't be any reaction because all reactions are passive forces.
- There is no active force, so you may not have passive force, that is no reaction.
- When there is no action,.
- There won't be any reaction.
- To use the equations of equilibrium, an appropriate free body diagram has to be written.
- To write the appropriate free body diagram, you should identify what are the forces acting, what are the supports that you get, and how many reactions are going to be acting at that support.
- See, taking all this into consideration, you have to draw the free body diagram.
- If you neglect that there is a joint at O, if you don't take these two, then there is chance of you making mistake.
- Support reactions at the supports.
- If it is an hinge, there will be two reactions, one vertical and one horizontal.
- If it is a built-in support, there will be three reactions, one vertical, one horizontal.
- Along with that, there is a moment also.
- If you make a mistake in writing the free body diagram, the entire solution that you are going to get will be wrong and you will not get the appropriate or the correct answer.
- in the examination as well as in the test.
- I hope all of you have understood these two problems clearly.
- We will stop today's class here.
- We will continue in the next class.
- That is in class number 26.
- We will try to solve few more problems where we will try to understand few other concepts what needs to be applied to solve such problems.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000134 · EV0000135 · EV0000136 · +1 source units in pack.json

10 UE26ME142A_Unit 2_Class 19.3_AV Summary

Study focus: UE26ME142A_Unit 2_Class 19.3_AV Summary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000138 · EV0000139

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

This is class number twenty six.

Key details

- My name is Dr.
- Venkatish from Civil Engineering under the Department of Science and Humanities, PS University.
- In the last class we had discussed few problems from unit two on equilibrium.
- We'll be continuing to solve few more problems in today's class.
- This is problem number three bar forty five from.
- The prescribed textbook, Engineering Mechanics Statics by Miriam Craig and Bolton, Volume 1, 8th edition.
- A person is performing slow arm curls with a 10 kg weight as indicated in the figure.
- The brachies muscle group consisting of the biceps and brachialis muscles is the major factor in this exercise.
- Determine the magnitude F of the brachialis muscle group force under the magnitude E of the elbow joint reaction at point E for the forearm position shown in the figure.
- Take the dimensions shown to locate the effective point of application of the two muscle groups.
- These points are 200 mm directly above E and 50 mm directly to the right of E.
- Include the effect of the 1.5 kg for arm mass with mass center at point G.
- State any assumptions.
- which is 350 mm.
- If I'm not wrong.
- The distance of G from.
- E can be taken at as 50 plus 100.
- At this point the force is going to be acting downward because it's the self fate of the forearm, 1.5 kgs into 9.81.
- Then he has asked you to determine the.
- Magnitude F and the magnitude E of the elbow joint.
- What is the magnitude of the resultant force acting at elbow joint E?
- And also what is the force acting in the muscle group.

- The muscle group could be taken to be acting along a line of action which is oriented in such a way that it is 200 mm above e point vertically, that is this one, and then it is 50 mm along the line EG.
- above E vertically by distance 200 mm.
- That line will be the line along which the muscle group force is going to be acting.
- That whatever may be the body that is acting, we are not interested in all that.
- What we are interested in is what is the magnitude of the force, what is the sense of the force, what is the line of action of the force, what is the direction of the force.
- What are the forces acting, what are the forces that we need to determine.
- What body is on which these forces are acting is not our concern.
- Our concern is only what are the characteristics of forces, where it is acting.
- What we need to find.
- That's all is what we are interested in.
- So if you look at the free body diagram, so we need to take the 10 kg weight that is acting because of the spear that the person is holding.
- Then 1.5 kg force, which is the sulfate of the forearm that is acting, 1.5 kg into 9.81 at point G.

Example or application from the lecture

- What you need to take from this statement is that the Chenkeji weight which is being held by the person in his hand, which will definitely be acting downwards, that distance is also given here.
- Then he also says that you take the weight of the forearm as 1.5 kg, which will be acting at point G, which is given here.
- Let's not worry about which muscle it is, why is that muscle coming, what for is that muscle, all that is not required for us.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000138 · EV0000139 · EV0000140 · +1 source units in pack.json

Core explanation · 05:00–09:59

Then the joint at the elbow, which will act as an hinge point.

Key details

- Then the force in the muzzle, as I have earlier explained, it is to the right of E at a point A, which is mentioned here, which is 50 mm from E horizontally, then above E vertically by a distance of 200 mm.
- The line joining this, that is from A point to this point, is the force, that is the sense of the force that's acting.
- The muscle is has to resist, so that's why.
- The arm will try to go down, the arm will try to go down, but that is been held in position because of that muscle and that is a line of action of the force if that is going to be acting.
- So first thing is find out what is the angle alpha made by the force F with the vertical as indicated here.

- Then as I told you, we are assuming that the force is going to be acting along this line, that is the assumption that we are making.
- So first thing is we need to find out what is the force acting at point E.
- To find out the forces acting at E, we need to find out what is the force F that is acting.
- Cause alpha and along the horizontal passing through E it will be F sin alpha, which will not produce any movement.
- Then 1.5 into 9.81 into 150, which will also be clockwise.
- That should be equal to the component of the fourth that is F, cos alpha vertically upwards, which will cross an anti-clockwise movement.
- That F cos alpha into 150 which will be anti-clockwise.
- Once you get the value of F, you can resolve it along vertical and horizontal.
- The horizontal component of the force is the only horizontal force that is acting on the body, but that has to be resisted by the reaction at E.
- Because your sigma f x is equal to 0.
- Then you take sigma f y is equal to 0, so you get e y to be equal to 1.5 into 9.81 plus 10 into 9.81 minus the component acting along the vertical because of the force f which will be f cos alpha.
- That's a component of the force acting in the muzzle, which will cause an anti-clockwise movement.
- Then four arm weight, which is 1.5 to 9.81 into the distance, the horizontal distance between the G point and E point, which is 150, and that would produce a clockwise movement, so negative.
- Then the self weight of the spear.
- Which is 10 into 9.81, the distance from E to the point where the weight of the spar is going to act is 350, that is it.
- That is what is indicated here.
- The sum of all this should be equal to 0.
- Next take sigma f x equals 0.
- Why is it negative?
- Because when I resolve the force F along the X it will go to the left.
- But right is the positive sense of x, so it is negative.
- Plus we have taken the force of EX to be acting to the right, so it is positive.
- Next takes summation of all the forces along vertical that is sigma f y equals zero, so component of the force F seven fifty three cost fourteen point zero four, acting vertically upwards, so positive.
- You have the force of the sphere acting downwards, the sulphate of the arm acting downwards.
- Plus we have a force acting upwards which is EY.

Example or application from the lecture

- They have also given you the x axis and y axis positive x and positive y.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000142 · EV0000143 · EV0000144 · +1 source units in pack.json

Core explanation · 10:00–14:59

I hope you have all of you have understood this problem.

Key details

- The device shown is used to test automobile engine valve springs.
- The dark wrench is directly connected to ROB.
- The specification for the automotive intake valve.
- Spring is that 370 newton of force should reduce its length from 50 mm, unstretched length to 48 mm.
- What is the corresponding reading here on the dark range?
- And what force F exerted on the torque range handle is required to produce this reading?
- Neglect the small effects of changes in the angular position of R O B.
- Clearly you can see that the spring will reduce its length from 50 mm to 42 mm.
- That's the difference in 8 mm.
- It will reduce or decrease in its length by 8 mm.
- When a force of 370 Newton is applied.
- Then he has also asked you to find out what is the corresponding force that needs to be applied.
- Movement is indicated in the ridge.
- Now you can if you observe here when I'm trying to apply this force, this would try to rotate because of which this would get compressed.
- When this has to get compressed, the force that has to be exerted on this is 370 newtons, and when I apply that 370 newtons, this would reduce the length by 8 mm.
- will be equal to whatever is the force acting at B into the distance.
- And it is going to be downwards.
- should be produced when we apply this force F.
- This would not produce any moment, that is the component acting along OA.
- which is F sine twenty will not produce any moment at O.
- Only the component F cos twenty which is acting normal to OA will produce a moment at O.
- Which will be equal to F cos 20 into 375.
- Oh, because of this 370 into 150 mm.
- At O is equal to 370.
- I told you that the force that's going to be acting on the spring is 370, but here the reaction force is taken, both of them are same.

Example or application from the lecture

- Next let's go to the next problem number three bar forty eight.
- Moment that is produced when you have this force and this much reduction is what is given.
- Then when you apply this force here, this angle that is given here as 55 will also slightly change because this would come down a little bit.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000146 · EV0000147 · EV0000148 · +1 source units in pack.json

Core explanation · 15:00–19:59

That should be produced by the F force that is acting.

Key details

- The movement produced by the force applied is F cos 20 degrees.
- into the distance between along OA is 375 mm converting it into meters.
- With a magnitude of 157.5, when I apply that, the moment produced at O will be equal to 55.5.
- Because of this, the spring would get shortened by a length of 8 mm, which is earlier was 50 mm.
- It will reduce to a length of 42 mm, that is it will get compressed by a length of 8 mm.
- I hope all of you have understood these two problems.
- These two problems that I have chosen.
- Are slightly difficult when compared to other problems.
- Get to know how to solve the other problems which are not solved, which are relatively easier than the problems that I have chosen so far.
- Those problems which are not solved, kindly solve it.
- And if you have any doubts in this,.
- In those problems, you can refer to the notes which is uploaded in Peso Academy, or you can approach me or any other faculty members in the Department of Civil Engineering.
- They will certainly be helping you out to understand how to solve those problems which we have not solved.
- That is class number 25.
- We'll be continuing to solve few more problems under equilibrium in the coming classes.
- I am Dr.
- Svangitesh from the Civil Engineering Department, under the Department of Science and Humanities, PS University.
- See you in my next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000150 · EV0000151 · EV0000152 · +1 source units in pack.json

11 UE26ME142A_Unit 2_Class 19.4_AV Summary

Study focus: UE26ME142A_Unit 2_Class 19.4_AV Summary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000154 · EV0000155

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Sv Venkatish from Civil Engineering, Department of Science and Humanities, PS University.
- This is class number twenty seven.
- In the last class we had discussed two problems on equilibrium and also I had informed that we'll be continuing in solving a few more problems before we go to the next topic in unit two.
- I've considered problem number three bar 49 from the prescribed textbook, which is Engineer Mechanics Statics by Miriam Craig and Borgan, Volume 1, 8th edition.
- The portable floor crane in the automotive shop is lifting a 100 kg engine for the position shown.
- Compute the magnitude of the force supported by the pin at C and the oil pressure P against the ATM diameter question of the cylindric of the sorry of the hydraulic cylinder.
- unit A B.
- Then you can see that there is a hook here to which there is some load that is there.
- And he says that it is hundred kg.
- For the position shown means this angle is 30 degrees, that is very clearly indicated along the line B C which goes and touches here.
- And there is a pin at C, so you will have two reactions here, one vertical and one horizontal.
- That is what is the support at Pen P, that is first thing.
- And also you can see that there is a cylinder here, which is the piston.
- Piston is there, then there is a oil filled in this cylinder.
- So when I apply this force that has been resisted by the by this piston which would try to compromise because of the oil present there, it is resisted by that, along with the support at C.
- What is the distance between B C along B C, and then what is the distance between B to the point where the load is applied along B C again?
- Then this piston is getting compressed, so internal forces will be moving away from each other.
- At B the force will be acting towards B from A.
- Then you will have two reactions here.
- If I take only this part separated out, then I have the pressure in the cylinder.
- Which will be acting towards B and if I take the force as F B, the line of action of that force which is acting at B has to be determined.
- Then we have this force hundred.

Example or application from the lecture

- So let's look at the next problem.
- For the angle that is given in the diagram, we need to find out what is the support at the ping C and also what is the pressure on the cylinder, hydraulic cylinder, the pressure on the oil pressure on the hydraulic cylinder is what we have to look at.

- They have clearly given that there is a crane here and you have wheels also which can be moved.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000154 · EV0000155 · EV0000156 · +1 source units in pack.json

Core explanation · 05:00–09:59

After determining what is the angle made by B C with the vertical or horizontal, preferably with the vertical, how can I determine this angle?

Key details

- That minus one fifty, I get the horizontal distance between A B.
- We already know what is the vertical distance between A C, I can also find out what is the vertical distance between C B, which would be four fifty sin thirty, which I'll get this one.
- Then I have this 100 kg force into the horizontal distance between them will be equal to 150, 1500, which is 1050 plus 450, which is 1500.
- Cross thirty degrees will be the horizontal distance.
- Then I can take algebraism of all the forces acting along the horizontal, so I get the horizontal reaction at C because the component of F P acting in the horizontal direction which will also be towards the right.
- Then the force acting because of F will be acting upwards.
- Then we have the force or the component of FP acting upwards, then we have this hundred kg which is acting downwards.
- Then once I have got the force in acting at B, F.
- That force divided by the area of the piston, I get the pressure acting in the hydraulic cylinder.
- I get the area.
- I hope all of you have understood this explanation.
- That is pi d square by 4.
- Multiply that with the force that is acting, so you get the pressure.
- Or you get the pressure into the force, we get the force, pressure into the area, we get the force.
- Or pressure is equal to.
- Force divided by area, that's what is indicated here.
- What is that?
- So this horizontal line and this line will be at thirty degrees because this is alternate angles.
- alpha, this total is alpha.
- I had considered this as alpha, but they have considered this as alpha, so time inverse of this distance by this, I have explained that how to get it.
- That is the vertical distance.
- The horizontal distance between this point to B.

Example or application from the lecture

- The area I can always determine because they have given the diameter of the piston.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000158 · EV0000159 · EV0000160 · +1 source units in pack.json

Core explanation · 10:00–14:59

Will be 450 which is this line.

Key details

- Cos of thirty which I get this distance total minus this one fifty, so I get the horizontal distance.
- And sine of this angle beta which will be the component perpendicular to C B into the distance 450 will give you the moment which is in the anticlocal direction.
- Next this force is 981.
- into what is the horizontal distance between C to this force, which will be one 150 plus 408 is 1500.
- Cos of 30, which I get this as the horizontal distance.
- That's what is indicated here.
- To be equal to zero.
- I get the force P which is acting in the piston, which we get it as three thousand nine hundred and twenty four newtons.
- Next find out the.
- Reaction at C, take all the forces acting along X axis.
- The force in the piston, cause of the angle, will give you CX.
- Because force in the piston is seven eighty one.
- Measure is seven eighty one.
- Then all the forces acting along the vertical, C Y acting upwards, plus nine eighty one which is acting downwards.
- If I take C Y to be acting downwards itself.
- That is positive.
- 981 acting downwards is also positive.
- The component of the force in or the piston is 3924.
- which is the force into sine of this angle.
- Then to find out the magnitude of C I will take the root of 937 square plus 2830 square.
- Resultant of the support reactions acting at CS two nine eight zero newtons.
- Hope everybody has understood this.
- This is a very interesting problem, slightly different than what we have solved so far.
- The statement of the problem was like this.
- During an engine test on the ground, the trust T is equal to three thousand newton is generated on the thousand eight hundred kg airplane.
- With masks center at G, the main wheels at B are locked and do not skip.
- The small tail wheel at A has no brake compute.
- the percentage change in N in the normal forces at A and B as compared with their engine of values.
- The self weight of the airplane is 1800 kgs and that is located or at B at point G.
- Then the main wheels at B are locked.

- Lock means the B wheels are not skidding, it won't skid.
- The small tailwheel at A has no brake, so this does not have brake.
- Whereas at this point, since there is.

Example or application from the lecture

- Let's move on to the next problem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000162 · EV0000163 · EV0000164 · +1 source units in pack.json

Core explanation · 15:00–19:59

Locked means it won't rotate.

Key details

- When the engine is switched on, when the engine is switched off.
- into the distance for point eight should be equal to 1800 into nine point eight one into this distance 0.8 so you get r n then take algebraic sum of all the forces along the vertical so you get r b here.
- When the engine is switched on, you will have in addition you have this thrust and also a reaction at B, horizontal reaction.
- acting along the horizontal.
- T is sine 12 acting along the vertical.
- Then I take some of all the forces along vertical.
- Reaction, horizontal reaction at B.
- Then of course I can find out what is the percentage variation acting at A and acting at B because of engine off and engine on position.
- The free body diagram, as I have told you, this is what it is.
- A point, then B point, you will have the normal reaction, NA, NB.
- Since this is locked, you will also have a horizontal reaction.
- Then the force acting at G and the engine when it is switched on you will have T equals three thousand metals.
- This is acting at twelve degrees.
- Next one engine of T equals 0, F B is equal to 0, that is the horizontal reaction of B will be 0.
- Taking sigma F is equal to 0, we get N A.
- Taking sigma MA equals 0.
- You have the engine switched on.
- There is this three thousand cross twelve degrees into the corresponding perpendicular distance should be equal to zero.
- N B dash that is the normal reaction acting at P.
- Then you take sigma F equals 0 normal reaction acting at E you can find out.
- So it is N A dash when the switched on condition, NG switched on, minus.
- Switched off condition, you have a switched off condition in 200 because percentage.

- So you get as thirty two point six and it is a negative because the reaction when the engine is switched on is lesser than the reaction when the engine is switched off.
- That you will get us or divided by hundred I should say, I'm sorry.
- This is a slightly a different problem than what we have solved earlier, but the procedure is the same except that you need to find out the percentage variation when the engine is switched on and switched off.
- So we'll be continuing to solve few more problems in the next class on equilibrium.
- My name is Dr.
- Sv Vengadesh from Civil Engineering under the Department of Science and Humanities, PS University.
- Kindly solve problems which I have not done.
- If you have any doubts, you can always contact me in the Department of Civil Engineering PS University.
- Meet you in the next class.

Example or application from the lecture

- So this could be resolved because they have given this as 12 degrees.
- Let's meet in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000166 · EV0000167 · EV0000168 · +1 source units in pack.json

12 UE26ME142A_Unit 2_Class 19.5_AV Summary

Study focus: UE26ME142A_Unit 2_Class 19.5_AV Summary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000170 · EV0000171

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Svinkitish from the Civil Engineering under the Department of Science and Humanities, PS University.
- This is class number twenty eight.
- We are in the process of solving few problems.
- Under chapter three, there is equilibrium chapter.
- And probably this would be the last class in this chapter.
- In the coming classes I will be starting with the new chapter, that is chapter number four, that is process.
- under equilibrium chapter.

- The authors are Miriam, Craig and Bulton.
- The statement of the problem goes something like this.
- The 50 kg point exerts a pull of 150 newton on the row.
- Rig the ashton.
- Compute the force supported by the pin at the hinge O.
- is hundred kg and we all know that the sulphate is always going to be acting at the center of the total span or total distance of the beam.
- The distance at which this 100 kg into 9.81 newtons will be acting is at a distance of 4 by 2, which is equal to 2 meters.
- And also he says that the weight of the boy who is standing on the beam is 50 kgs.
- It will go to act at the center of gravity of the boil, which is indicated in the diagram as 1.5 meters from the left support, that is this distance where the weight of this person or the boy is gonna act that is fifteen to nine point eight one kgs will be acting at this point.
- And it also says that the boy exerts a pull of 150 newtons.
- Whereas at this point it will be acting upwards.
- Since the boy is standing on the beam, the weight of the boy is going to be acting on the beam.
- Whereas when he tries to pull this rope, the tensile force always going to be in pair, acting towards each other.
- However there will be force in this cable also.
- Again here it will be opposed to this which will be acting.
- Like this, from moving away from here, here it will be moving away.
- We will have a force here which will be acting downwards.
- We will have a force here which will be acting downwards.
- In addition to this, we will also have a roller here at point A.
- Then we have a pin or an inch connection at O.
- What are the forces that's going to be acting on this beam now as two reaction at O, one vertical, one horizontal, one reaction at A which will be acting up vertically upwards.
- We have the.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000170 · EV0000171 · EV0000172 · +1 source units in pack.json

Core explanation · 05:00–09:59

Tension force in this cable which is acting downwards of magnitude 150 newtons.

Key details

- Here also there will be one more cable which will be acting downwards of magnitude 150 newtons, and we will have the weight of the which is fifteen to nine point eight one.
- In addition we will have also hundred into nine point eight one will be active in the center of the beam.
- The forces in this cable will balance with each other, remember this carefully.
- Or what is that we have to find out?
- We have to compute the force supported by the pin at the end.

- that is O.
- You take the momentum out EA.
- Or movement about O because that's where you have the unknown support reaction that is acting.
- If you take the moment about A, you get the reaction acting at O, vertical reaction into 0.75, I assume it to be acting upwards, it will be anti-clockwise.
- Then the sulfate of the beam which will be acting.
- But the center of magnitude hundred into nine point eight one.
- But from A it is going to be at a distance of 1.05.
- Then the force in this will also act in the anticlockwise moment, which will be equal to 115 to 2.5.
- Then we will have the force that is acting in this cable which is also equal to 150 newton into force is 3.25.
- In addition we will also have the weight of the boil which is 15 to 9.81.
- From this end if I want to find out it will be equal to 3.25 minus 1.5, which is going to be 1.75.
- And that vertical reaction we will find out.
- free body diagram I'm not sure it is not visible here, however I have told you.
- That is at the point where the roller is acting, as I have told you, and one fifty is the force in the cable into 3.25 that is the extreme end left cable, then one more cable which is acting 1.15 into 2.5 plus the sulphate of the beam.
- Should be acting at the center from A it is 1.25 plus the weight of the boy which is 15 to 9.8 one acting at it.
- All this are clockwise, so it is taken as positive minus the reaction acting at the hinge into the distance which is 3.25.
- However, if you take clockwise as positive and anti-clockwise as negative, or clockwise as negative and anti-clockwise as positive, you get the same equation.
- And that's what I've told you.
- Force in the cable is not considered because that will be acting opposed to each other, that will balance with each other.
- And also there is no other horizontal reaction that is acting, that's why the reaction at.
- O along the horizontal direction is a zero.
- So we move on to the next problem.
- Next problem is 3 bar 54.
- The pin A which connects the 200 kg steel beam with center of mass at the G to the vertical column is welded both to the beam and to the column.
- To test the weld.
- The eighty kg man loads the beam by exerting a eighty newton force on the row, which passes through a hole in the beam as shown, calculate the chart.
- or couple yum supported by the pin.

Example or application from the lecture

- Let's move on.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000174 · EV0000175 · EV0000176 · +1 source units in pack.json

Core explanation · 10:00–14:59

There is a pin at here, that is this one.

Key details

- Then he says that the beam is welded.
- To the column.
- To test this weld, because without the weld, the beam would try to rotate in the clockwise direction.
- Since there is a weld there.
- This would resist the rotation.
- And he also says that the person exerts a 800 newton force on the rope, which is passing through the beam by a hole, which is drilled.
- The same rope is continued here.
- Here also you will have the 800 newton force.
- And that is taken by this pulley.
- But however if I cut it here, there will be 1804 newton which will be acting downwards, plus here also.
- Because of this is asking calculate the target M or the couple M.
- Supported by the pin that is at this point what is the moment that is being supported or the target that is being supported.
- You have the reaction R here.
- The horizontal reaction is not indicated because there is no horizontal force here.
- Then you have a momentum that is indicated in the anticous direction.
- Cell weight of the beam, cell feed of the person, then the cable weight.
- Which will be acting on the person itself, even though there is a hole there, it is not going to be directly acting on the beam.
- It is going to be transferred from the.
- Plus, in addition to that, you will have also the cable which is attached to this point, so you will also have that force.
- The force acting is.
- The sulfate of the bee?
- Self rate of the person.
- As I've explained.
- We get the s first sulfate of the person eighteen to nine point eight one.
- The distance at which it is acting is thousand eight hundred from the left end, that is A.
- Plus we have one hundred.
- Kg that is 29.81 into thousand two hundred is the sulfate of the beam that is taken here plus the cable whatever is there that has been taken here.
- Only one unknown is M and this is acting in the.
- anti-clockwise direction, all the clockwise forces or the moments are taken positive, anti-clockwise moment is taken as negative.

- I hope all of you have got this.
- I have a doubt of whether the force that is acting.
- Or the force with which the person is pulling is not 800, I think it is 300.
- Make that correction, please.
- It is 300 newtons.
- No, sorry, it is kilo Newton meetings.

Example or application from the lecture

- The sulphate of the beam is given as two hundred kgs, which will be acting at the center, that is given as G here, which is at a distance of thousand two hundred mm from the left hand.
- You can see here, they have given that it is welded here.
- And they have also given you the weight of the person which is ADKG, which will be acting along the center line of the person, which is at a distance from A by 1200 plus 600, which is 1800.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000178 · EV0000179 · EV0000180 · +1 source units in pack.json

Core explanation · 15:00–19:59

We have come to the end of equilibrium chapter.

Key details

- So basically what we have to understand in equilibrium chapter is first thing is the equations of equilibrium.
- For a two dimensional there are three equations of equilibrium.
- One is sigma.
- F along the vertical, algebraic sum of all the forces acting along the vertical is zero.
- Algebraic sum of all the forces acting along the horizontal is zero, then algebraic sum of the moment at any point should be equal to zero.
- They are called as determinate structure.
- If you have more than three constraints,.
- Then the number of constraints more than three is called as a redundant reaction or constraint.
- And we need to have more equations to solve such problems other than the three equations of economy.
- Such structures are called as indeterminate structure.
- Then we should also understand what are the different types of supports that we have.
- Smooth surface, rough surface, roller surface, or roller support, then sliding guide, then pin connection, building connection, gravitational force, linear, spring,.
- And torsional spring.
- Then how to draw the free body diagram.
- Then what are the different types of supports that we have?
- How do we replace these supports by reaction.

- With this we will be able to solve the problems in case of equilibrium chapter.
- In the next class, I'll be starting with chapter four, which is on trusses.
- Again, in trusses, we will be continuing to use these equations of equilibrium along with what we have learnt in unit one, that is resolution of forces.
- Moment topic, couple as well as the resultant.
- Knowledge is very much required to understand chapter four, that is process chapter.
- Okay, along with that you should also understand other things related to trusses which I will be discussing more in depth when I go to take it in the next class.
- Whatever problems I have discussed here, there are other problems also which I would request you to go through and solve them.
- I'll be very happy to help you.
- We have come to the end of today's class as well as end of chapter three, that is equilibrium.
- My name is Dr.
- Sankitesh from Civil Engineering under the Department of Science and Humanities.
- That is class number thirty.
- However, we have one more class that is 29, which is kept for experiential learning.

Example or application from the lecture

- I have explained this, which is given also in the textbook on page number one hundred and eleven if I'm not wrong, and as well as hundred and twelve, where we have discussed about cable.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000182 · EV0000183 · EV0000184 · +1 source units in pack.json

13 UE26ME142A_Unit 2_Class 21_AV Summary

Study focus: UE26ME142A_Unit 2_Class 21_AV Summary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000186 · EV0000187

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- SV Vengadesh from Civil Engineering under the Department of Science and Humanities, PS University.
- This is class number thirty-one.
- We are in the process of understanding.
- Chapter 4, that is trusses under unit 2.

- In the last class I had explained about what is a truss, what is trust made up of, what are the different types of trusses that we have, what is plain truss, what is.
- The load that can be applied on trusses as well as what is tension forces acting on the members and what is compressive forces acting on the members.
- And also I talked about what are the different methods of determining the forces in the members.
- which is either in going to be in the compression or at tension.
- which explains what are the different methods of determining the forces in the members.
- The first method is method of joint.
- There's one more method which is called as method of section, but in the syllabus prescribed for you, we have only method of joint, so we'll concentrate only on method of joints.
- Since there is no horizontal force acting on the truss, there is no horizontal reaction acting at A.
- Now coming to method of joints, the first step that is involved in method of joint is to identify a joint which has got only two unknowns.
- None of the joints have got only two unknowns unless and until we determine the support reaction R1 and R2.
- If we determine the support reaction R1 and R2, there will be two joints that is joint A and joint D which will have only two unknowns.
- That is in member AF and member AB as well as if you take joint D it will be member DC and DF.
- However, here we have considered a joint A, that is this joint A.
- is the external force, which is the reaction force, that is R1, and then there are two members, AF and AB.
- I am assuming the force in the member A B as tension.
- This is external force, this is internal force.
- Next the force in A the force in AB is considered to be moving away from the joint, which is under tension.
- Whereas the internal forces will be acting away from each other.
- This is the external force is acting towards there will be one more here.
- Internal forces is acting away from each other.
- So that means it will be towards this joint.
- AB that is this angle.
- I can resolve the force in the member AF along vertical as well as along horizontal and then use the equations of equilibrium to determine the forces in the member AF and AB.
- There are two equations of equilibrium, so we can determine these two forces.
- Next after taking this joint A, next we have to choose the other joint which has got only two unknowns.
- Again, we can resolve the force in the member AF along horizontal and vertical, use the equations of equilibrium to find out the forces in F E and F.
- Once I determine these two members, now I can take joint C, I can determine the forces in these two members.

Example or application from the lecture

- Now to explain the method of joints I must take a simple example as plane trust ABCDEF At A we have a pin connection, at D we have a roller connection and there is a force applied at B which is here which is also vertically downwards.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000186 · EV0000187 · EV0000188 · +1 source units in pack.json

Core explanation · 05:00–09:59

Then I can take joint E where I will be knowing all these three members.

Key details

- There will be only one non-on which is E D which I can solve.
- Then if I take joint D, now we know the force in the member E D as well as the force in the member C D.
- And if I had already determined R2, I would also be knowing R2.
- But what we could do here is we can take R2 as an unknown.
- Determine R2 by using the forces that we have got from the member ED and C D, and check whether we get the same answer, whatever we would have got earlier.
- in the members.
- Now in the entire engineering mechanics, whatever that is included in the syllabus, only for trusses we would be able to say whether the forces that we have got in the members are correct or not because we have a chuck in this case.
- That is under tension.
- Then if you take the member F B, again the forces are acting towards each other, that is under compression.
- Then if you take FE, the forces are acting away from each other is under tension.
- This is the method that we should use to get the forces in the members.
- I hope all of you have understood.
- The first thing is you have a member A B C D okay like this, it's a square one.
- And then we also have the member.
- B C and AB.
- This is just overlapping each other and each of them can move individually without the joint considering them.
- Now if because of the external forces, if the body is struck like this,.
- And also if you observe at each of the joint there are more than three members.
- Here these diagonal members are capable of supporting only tension.
- It cannot support compression.
- This member AB and D are cables which can take tension which cannot take compression cable or row.
- You can easily identify that this diagonal T C is getting compressed, whereas this diagonal AB is getting elongated.
- So because of which I can say that.
- The force in the member A B will be under tension, whereas the force in the member D C will be under compression.

- Then I can take this joint since I have no force in this, I can find out the force in these two, and similarly I can go out and join the other joints also.
- Then now coming to other special cases if there are only three members of the joint and if two members are.

Example or application from the lecture

- All the joints are given here, and then finally we will find out the forces in all the members, and once we get the forces in all the members, then we will represent those forces on the truss like this.
- For example, if you take AF, since the forces are acting away from each other, they are internal forces.

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EV0000190 · EV0000191 · EV0000192 · +1 source units in pack.json

Core explanation · 10:00–14:59

Collinear, okay, the forces in these two members are going to be equal because of which the force in this member when I resolve it, if I take some of all the forces along this, the force in this member will be simply zero.

Key details

- Only two members of the joint and they are acting like this, then if you take any axis x and x dash, the forces in those two members will be zero.
- Then if you have a crossing of a members, since this member is collinear, the member in these two four.
- forces in these two members will be same similarly the forces in these two members is going to be same it is not essential that they have to be at retire decrease only so if the members are collinear if they are in a straight line then the forces in those two members is going to be same okay.
- Problem 4 bar 5 is what is taken here.
- And you can also look at those joints.
- At joint A, there are only two members since we know the load at A already, so there are only two unknown forces that is acting.
- And the force acting in this member is 4 kilometers.
- I think that is missing here.
- Take that force acting at joint A as 4 kilometers.
- Angle BAE which is going to be this angle.
- Why am I doing this?
- Because it is easier to say initially that the forces are moving away from the joint.
- And once I get the values of this, if it if my assumption is correct, I get a positive answer.
- If my assumption is wrong, I'll get a negative answer, which indicates that the force in that member is a compressive force.
- This one, this force in this member AF is acting away from A.
- I get the force in this member F A B.

- This is forty five, so sine forty five which is acting upwards.
- And the force, the external force which is acting downwards is 4 kilonewtons.
- Next, I'll take the other equation of the equilibrium that is summation of all the forces, algebraic sum of all the forces along X equals 0.
- AB cos 45 which is acting towards the right so positive.
- We will keep the sign positive or negative as it is.
- When we are substituting somewhere else for the value of this member forces A.
- I will take the sign also into consideration as it is as negative.
- Or if it is positive, I will take it as positive.
- Because I am assuming the forces in all the members as tension itself that is moving away from each other, it will help me out very much.
- Next I will take joint E.
- Then if I take sigma f x is equal to zero.
- we get the force in the member E D to be same as the force in the member.
- AE or EA, both of them are same, which is also going to be equal to when I substitute the value of minus 4, I get this to be minus 4.
- Next I will move on to joint B.

Example or application from the lecture

- This is the trust that is given.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000194 · EV0000195 · EV0000196 · +1 source units in pack.json

Core explanation · 15:00–19:59

FAB with a negative sign.

Key details

- Next, I'll take summation of all the forces along X is equal to 0.
- If I solve the joint E I will get E B.
- Then if I solve for joint B I will get the force in member B D.
- Okay now using the special conditions I can initially say that the force in the member BE is equal to zero because there is no force acting at E.
- And these two member that is A and E D are collinear, the forces in those two members are going to be equal.
- We have got the force in the member BE as zero, okay because there is no force acting at E and B D as minus five point six kilometer which indicates that it is under compression.
- I hope all of you have understood this.
- I've considered four.

Example or application from the lecture

- Let's move on to the next problem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000198 · EV0000199 · EV0000200 · +1 source units in pack.json

14 UE26ME142A_Unit 2_Class 22_AV Summary

Study focus: UE26ME142A_Unit 2_Class 22_AV Summary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000202 · EV0000203

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr Svinkitesh, a civil engineer from the Department of Science and Humanities, PS University.

Key details

- in the last class we had discussed about unit two equilibrium chapter and also we had discussed few problems on equilibrium chapter.
- Today we will be starting with the next chapter, that is chapter number four.
- from the textbook which is on trusses under unit two as prescribed by the university PS University.
- A truss is a structural member which actually carries the load from the roofing element or a slab or a deck increase of a bridge.
- And transfers the load coming on this roofing elements to the column on either side and finally this load will get transferred to the foundation.
- From foundation it will get transferred to the ground below.
- The stringers are supported by this rafters or cross beams as they are called.
- And this cross beams are attached to this structure, which you can see on either side of the bridge.
- There is one here and there's one more here.
- Which are nothing but the actual truss.
- The load from the vehicle and the Dex Lab will transfer to this stranger.
- Then the string from stranger it will get transferred to the cross vehicle.
- Wherever there are joints.
- Here also you will have a load L.
- to each other and you will have the vehicle moving here and in case if it is a roof truss, usually the roof trusses are going to be slope.
- Then from the purlin or the cross staff, it will get transferred to the trusses on either side.
- A truss is basically used when the span between the supports.
- There will be another half on the left side of the screen.
- And you will have a support on either the side.
- Then we go for such type of structure.

- And also when we are not intended to construct an upper floor above the truss, we go for truss.
- And usually the truss will be constructed of either wood.
- Earlier we used to do it in wood, but nowadays since it is very difficult to get wood and also there is restriction from the government for the use of wood.
- we have moved on to using iron or steel.
- It can also be manufactured using aluminium or any other suitable metal okay for the purpose.

Example or application from the lecture

- This is the Dex Lab and you can see this, they have given it as strangers, okay.
- From cross beam it will get transferred to the truss on either side, and you can see they have given the load at the joint, which is given as L.

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EV0000202 · EV0000203 · EV0000204 · +1 source units in pack.json

Core explanation · 05:00–09:59

Constructed of a lighter material, lighter weight material.

Key details

- Vehicles that will be moving on this bridge.
- That weight will be quite large when compared to the self weight of this truss itself.
- The in the problem if it is mentioned to be considered the sulphate, we have to take the sulphate into account.
- Bridge trusses that are employed.
- This case you have a central member and of course it is symmetrical when you see the geometry of the truss, but you can see the inclined members which are moving from top to bottom.
- from left to right.
- And you have four members here.
- Members are the structure which are in between two joints.
- Likewise you have several member here.
- Ends of the member will be will be connected to what is called as a joint.
- Unless you understand a joint, you cannot understand member.
- A member is a structure which is in between two joints.
- But this is a joint where you have two members meeting.
- Plus, of course, you have the support reactions also here.
- This is also a joint where you have three members meeting.
- If you see the left side, the inclined members are moving from left to right, okay, and it is upwards, it's moving upwards.
- Whereas here, except the end, the other was moving from right to left, from up to down, okay.

- And the right side of the members, the income members are moving from top to bottom from right to left.
- Each member moving in a zigzag way like this, the inclined member.
- One member if it is from left to right, the other member is from right to left, top to bottom, and this is from bottom to top.
- And this is a K member you can see that this looks like a K joint, okay?
- Here also it is inverted K joint.
- That is named after the person who has invented this.
- It is based on how you connect the members and how do you take the joints into account.
- One more thing that you should remember in case of processes.
- The members in the trusses will be loaded at the joints only.
- None of the member will be loaded in between.
- That is because.
- The members of a truss can take either a compressive load or a tensile load only.
- Next, coming to what are the different types of roof trusses that you have, there are four different types that is mentioned here.

Example or application from the lecture

- For example, if you take the weight of the.
- With this background, let's now look at what are the different types of.
- So the first five which is given here are the trusses for bridges.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000206 · EV0000207 · EV0000208 · +1 source units in pack.json

Core explanation · 10:00–14:59

One is Fink, then it is Pratt.

Key details

- Again Hovi and Warner.
- Here you have.
- Member from going from the incline members going from bottom to top, from left to right, whereas in this case it is moving from top to bottom from left to right, the left side.
- Whereas the right side, you can see here it is the incline members are moving from.
- Top to bottom from left to right, whereas here it is bottom to top from left to right.
- And of course they are symmetrical, all of them are symmetrical here.
- Now you have to remember these type of trusses because in the examination, ISA or ESA they can ask you to draw a.
- A Pratt trust or a hobby trust, or they can give you a trust and ask you to name the type of trust.
- A plane truss is a truss in which you have all the members and the joints in the same plane.
- If you take it, it is in the horizontal plane only.
- As I have told you that the members of a truss is made up of either only compression or only tension.

- Tension and compression will have a pair of forces, either they will be moving away from each other or moving towards.
- If the external forces are moving away from each other, the body would tend to elongate.
- Here it will be moves towards this joint.
- Here it will be moving towards this joint.
- Such forces are defined as tensile force.
- Comping coming to compressive force.
- If the external forces are moving towards each other along the longitudinal direction of the member, the member will try to get compressed.
- Such members are called as compressive members.
- The bottom half and take it separately, you can see here.
- This is external force, compressive force, C, the internal force will be acting towards the joint for this half of it.
- Whereas if you take the upper half, the external the internal force will be moving towards this.
- The internal forces that is away from each other, the external forces will be moving towards each other, the body will be under compression.
- In fact, I am going to repeat this again, taking the tension into consideration.
- That is what is called as tensor member.
- Now, because of the external forces, if I take the internal forces, the internal forces will be acting towards each other, that is towards the joint.
- The body would get elongated, so it is under tension.

Example or application from the lecture

- For example, if you take here, you have three members here, all these three members.
- So here is an example of what is tension and what is compression.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000210 · EV0000211 · EV0000212 · +1 source units in pack.json

Core explanation · 15:00–19:59

Now, what because of what these forces are coming?

Key details

- Because of the external force that is acting at the joint.
- And as I told you, in case of trusses, the lore cannot be applied in between the members.
- It has to be applied at the joint only.
- These are external forces.
- Because of these external forces, you get the forces in the members which are internal.
- This one also is under compression, whereas this one is moving away from the joint.
- Kindly remember this because this is going to be helpful when we are going to be solving problems on trusses.
- And especially in case of trusses.

- What we are going to be doing is to determine what is what are the forces that is acting in each of the member and what is the type of force that is acting, whether it is compressive or tensile.
- And how do we determine the forces in the members?
- There are two methods.
- First method is method of joints, the second method is method of section.
- Now your syllabus you have only method of joints and that is what we are going to be concentrating on.
- The end A is supported with a hinge support or a pin support, so you have two reactions, one vertical and one horizontal, and end B is supported with a roller support, so you have one reaction which will be acting vertically upwards.
- We have to first determine the support reactions.
- In this case, to identify a joint which will have the number of unknowns or the number of members where the unknowns are to be determined, which are the member forces, which are only two members are there.
- So in case of method of joints, what we do is we take each joint separately as indicated here.
- You can see here they have indicated the joint A here.
- No other forces are acting at any of the joint.
- The reactions are also going to be vertical.
- And they have written the diagram here.
- Considering the joint A only separately here.
- This is joint A where you have a reaction at A acting which is written as R.
- Then you have the member AF as well as the member AB.
- The member EAF, it is considered to be under compression.
- That means forces will be acting towards the joint.
- You can see that is what is considered here.
- See here the member internal forces is acting towards the external forces will be acting away.
- Whereas the member A B it is considered to be in tension, so that means the forces will be acting away from the joint.
- This is the.
- Force that is reacting external force.
- That is algebraic sum of all the forces along X to be equal to 0 and algebraic force of all the forces along Y to be equal to 0.
- I can determine the forces in the member AF and AB.
- Once I find the forces in these two members, I can next go to the next joint, which will have only two unknowns.
- That is, the next joint is A is F after A.
- I am taking the next joint as F.
- The m force in the memory AF is already determined from this joint.
- The sense of the force in this is moving towards the joint again.

Example or application from the lecture

- Now there's a simple example that is given here, a roof thrust.
- So we have an example here which is a roofing truss that is taken.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000214 · EV0000215 · EV0000216 · +1 source units in pack.json

Core explanation · 20:00–24:59

This one AB will not come into picture when I take the joint Ef, you can see here this is F.

Key details

- Then we will analyze this again by using the equations of equilibrium to find out the these two unknowns, okay, which we can determine by using sigma fx equals 0 and f y equals 0.
- Once I find this, so now we have found out these three members along with this.
- The external force that is applied is also known.
- Then you can take the next joint as.
- Joint C here.
- This member is known.
- These two I can find out.
- The mem force in this member will be zero.
- We will look at that a little bit later.
- Then I can take the force in the in the joint E.
- We have already determined the force in this, that is zero member in this.
- Then coming to this joint D, here we would have determined these two earlier itself, and also R2, but we can take R2 as an unknown.
- from these two forces whatever we know, if I can find out R two and I can compare what is the value that I have got earlier to what I will get here.
- See you in the next.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000218 · EV0000219 · EV0000220 · +1 source units in pack.json

15 UE26ME142A_Unit 2_Class 23_AV Summary

Study focus: UE26ME142A_Unit 2_Class 23_AV Summary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000222 · EV0000223

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Sv Vength, Civil Engineering, Department of Science and Humanities, PS University.
- This is class number thirty-two.

- We are in the process of solving some problems.
- In chapter four, unit two, that is on trusses.
- In the last class we had solved one problem.
- Today we will be continuing and solving some more problems.
- This is I've taken problem number four bar eight.
- The statement of the problem goes like this.
- Determine the force in each member of the loaded truss.
- You have load at A, which is twelve kilonewtons, and load at E which is eighteen kilonewtons.
- B D and B C as well as C D.
- So to find out the forces in this member, first thing that we have to do is determine the angle made by the member AB with the horizontal that is angle BAE.
- which we will get it using this triangle ABE.
- Sorry about this should be tan theta 2.
- Because we are assuming it to be under tension, internal forces in case of tension will be acting towards each other.
- Sine of angle theta 1 should be equal to 12 kilonewtons, that is what is mentioned here.
- Next we take some of all the forces along the x direction.
- So we get Fe to be equal to minus 19.2 kilovattan, which means that it is under compression.
- However we will keep the sign as it is as minus and take that force to.
- B in the member A E to be in tension only.
- Next coming to joint E.
- If I take joint E, the force in the member E B will be equal to 18 kilonewtons, that's what is mentioned here, which will be under tension itself.
- Then you take summation of all the forces along X.
- We have got AE as negative, so the force in the member E D will also be negative, which indicates that it is under compression, which will be of the magnitude nineteen point two itself here also.
- Next we will take the joint B.
- We have determined these two joints already, we know the these two members, sorry, not the joint.
- The forces in these two members is known, so we can find out the force in these two members.

Example or application from the lecture

- Consider all the members to be under tension, that means we are taking the forces in each of the member moving away from the joint.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000222 · EV0000223 · EV0000224 · +1 source units in pack.json

Core explanation · 05:00–09:59

as well as along horizontal.

Key details

- Sigma F y equals 0.
- Whereas this would be negative, so minus FP.
- B D also will be negative minus F D sine twenty two point six one which is this angle.
- Then we have the force in the member BA, which will also be negative, which is FAB sin 32.
- That is what is indicated here and also the component of the force B a along x.
- BA along x will be negative, so that is what is indicated here.
- Whereas this two is going to be positive because of these two forces.
- F B D in this equation, we get the force in the member B C which is positive sixty five point nine four kilonewton.
- So that means.
- The force in B C will be tensile force of magnitude 65.94.
- Substituting this value back in this equation, we will get the force in the member Bd as minus 50.05 kilonewton.
- Only one member that is left is this member that is C D.
- And we can we know the force in this member, this member.
- some of the forces that we have got it as positive, some of the forces we have got it as negative.
- You can indicate the force in the member if it is tensile, it is acting towards the joint.
- It is acting away from the joint.
- Or there is another way of indicating the forces in the members with respect to a table.
- That table is going to be looking something like this, where you will have serial number, member.
- That is the name of the member and what is the force magnitude that is acting in that member and what is the type of force that is acting, whether it is compressive or tension.
- A B A E B A E Oh there is a mis equal to.
- AE is equal to E D, I think this should be E D.
- Then we have B C and B D and C D, and corresponding values of the forces that we have got it to while we have analysed it is what is mentioned along with the sign.
- If it is negative, we will write it as compression.
- Or why should we indicate whether it is tension and compression?

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000226 · EV0000227 · EV0000228 · +1 source units in pack.json

Core explanation · 10:00–14:59

That determination is different for a compression member and it is different for a tension member.

Key details

- That's why we should know whether the member is under compression or tension.
- However, in the first year syllabus, which is common for all engineering branches, we just have to determine what is the magnitude of the force and what is the type of the force.
- We will be stopping at this point.
- Further design of the members will be taken by the civil engineering in their IR submission.
- This problem is problem number four bar ten.

- Again, we have taken these problems from the prescribed textbook that is Engineer Mechanics Statics by Miriam and Craig and adopted by Colden, 8th edition only, or eighth volume.
- Or eighth edition only is what we have adopted.
- The problem statement goes something like this.
- Determine the force in each member of the loaded truss.
- Make use of the symmetry of the truss and of the loading.
- What is that hint?
- It says that make use of the symmetry of the truss and the loading.
- What does this mean?
- No, we say that a trust is symmetrical.
- Three conditions are satisfied.
- What are these three conditions?
- The three conditions is if you identify a truss having an symmetric axis, then the truss should be symmetric in.
- Geometry to the left of the section as well as to the right of the section.
- As to the light, right?
- That is the first condition.
- The second condition is the loading also should be symmetrical with respect to the symmetric axis.
- That means we have a loading at turn.
- As 30 kilometer, which is to the left of the symmetric axis Cg, which is at a distance of 5 meters from G.
- However, we have a loading at G itself and that is along the symmetric axis itself, so it is symmetrical.
- Okay, so two conditions one is symmetry with respect to geometry, the second condition is symmetry with respect to the loading.
- The last condition is it should be symmetrical with respect to the reactions also.
- Note, I'm telling it should be symmetrical with respect to reactions also.
- What is this?
- We get the reaction because of the supports.
- One vertical and one horizontal is what we take, but E we take it as one reaction which will be perpendicular to the line of the roller, which is in this case is line of roller is horizontal, so your reaction will be acting vertical.
- If you absorb the entire truss, we don't have any horizontal load acting external.
- Therefore, horizontal reaction at A is all zero.
- Since the horizontal reaction at A is zero, we will have only vertical reaction at A as well as vertical reaction at E.
- And this is symmetrical with respect to the symmetric axis.
- That means from G the reaction or the load at A is at a distance of ten meters.
- The advantage of a symmetrical truss is that.
- Whatever forces that we get to the left of the symmetric axis in the members, in all the members, the same forces will be acting to the right side also.
- That is the advantage.

Example or application from the lecture

- Let's go to the next problem.
- We have to find out the forces in all the members, but he has given you an hint.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000230 · EV0000231 · EV0000232 · +1 source units in pack.json

Core explanation · 15:00–19:59

If I could find out the forces in this six members, one, two, three, four, five, seven members.

Key details

- One, two, three, four, five, six and seven members, the forces in the member GF will be equal to GH.
- CF will be equal to CH, C D will be equal to C B, then your DE will be equal to PA, then your F E will be equal to H G.
- The next thing.
- How do we solve this?
- That's explanation for a symmetrical process.
- As for the method of joints, we have to identify a joint which has got only two unknowns.
- But if you observe here, we cannot find out any.
- Joint which has got only two unknowns, there are more than two unodes.
- But if I could somehow determine the reaction at A and reaction at E, then I can start with joint A or joint E.
- The vertical reactions will also be symmetrical.
- Therefore, I can easily say that reaction at A is equal to reaction at E, which will be equal to total vertical forces that is acting divided by 2.
- What is the total vertical force that is acting?
- That's why we get the reaction at A along the vertical and the reaction at E to be equal to 60 kilowattons.
- And as I have already explained, if you take sigma f y equals 0, you get the reaction at.
- AE at that is the horizontal reaction at A will be equal to zero.
- Next thing is we have to determine what is the angle made by the member AB with respect to AH or the horizontal, that is the angle PAH, which we can take this triangle and find out this angle, which I if I take it as theta, then it will be done inverse of 4 by 3 which is indicated here and that angle will be 38 point so.
- Compressive and the force in the member AH as tensile, then that's why you can see here force in the member AH is moving away from the joint, force in the member AB is moving towards the joint.
- Sigma F equals 0, sigma f x equals 0.
- If I take sigma f x equals 0, I will get the force in the member AH.
- You can look at the value of this in the PPT that has been attached in PSO Academy.
- So continuing this, next we will take joint B.
- In joint B again, we have assumed the force in the member.

- A B as compressive, so here also we will be assuming the same thing to be moving toward the joint.
- Whereas at B the force is moving away from the joint, we are assuming it to be under tension, this also the force in B H also as tensile force.
- We had got the value of the force in the member AB as 96 kilonewton, which is compressive, and that is positive, we have got.
- And then now we will apply the apply the equations of equilibrium after resolving this 96 kilometer force along vertical and horizontal.
- Next you take joint H in joint H.
- We have got the value of P C which is seventy-five.
- We have got it as negative.
- Then 60 kilon is the force acting upwards, 30 kilonewton is acting downwards.
- How did we get this 60?
- We have got it as BHS sixty tensile, so that is as it is.
- Then these two are to be determined.
- I get the value of CH and GH and this is what the values that we have got.
- Then CG is assumed to be tensile, we get it as positive and that is why it is tensile.
- Next you can take joint at G.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000234 · EV0000235 · EV0000236 · +1 source units in pack.json

Core explanation · 20:00–24:59

Joint G we have four members.

Key details

- And it will be under tension.
- We will continue in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000238 · EV0000239 · EV0000240 · +1 source units in pack.json

16 UE26ME142A_Unit 2_Class 24_AV Summary

Study focus: UE26ME142A_Unit 2_Class 24_AV Summary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000242 · EV0000243

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Sv Venkitish from Civil Engineering under the Department of Science and Humanities, PS University.
- This is class number 33.
- In the previous classes we had discussed about chapter four, which is on trusses under unit two.
- We had solved a few problems, so we'll be continuing solving a few more problems in today's class.
- The statement of the problem goes like this determine the forces in members A B, B C and B D.
- of the loaded truss.
- Why it is not symmetric?
- Because we have a horizontal force here at D.
- And if you take the symmetric axis as B D.
- Of course it is symmetric with respect to geometry because you have the same number of members to the left as well as the two right of this line B.
- However, the loading you have only at T which is acting towards the right, we don't have a loading towards the left.
- Supports that we have here is an hip support and a roller support.
- So since there is one vertical and one horizontal, we will also have a horizontal reaction because we have an horizontal force there, yeah.
- Next, we have to determine the forces in few of the members, whatever is asked according to the problem.
- We will first have to determine the support reactions because we cannot start with any of the joints, which has got only two unknowns, where we can determine the forces.
- So we need to first find out the support reactions.
- Which is also equal to L by 2.
- Then if you take the algebraic sum of all the forces along X direction, we get the reaction at A, which will be acting towards the left, which is equal to L.
- Next we can take any of the joint, either joint C or A.
- The force in the current member will be or should be taken to be same as the force in the member where you join that two points or the joints of this member.
- Find out what is the force that is going to be acting along that line C D.
- The same will be the force that is going to be acting in the current member C D as well.
- So the angle made by DAB will be 45 degrees because this is also R, this is also R so I can resolve the force in the member A D along the vertical and horizontal and use the equations of equilibrium.
- Then you take the force in the sorry you will have the.
- I am sorry about that.
- First you resolve this force along vertical and horizontal.
- You take the algebraic sum of all the forces along vertical.
- Sine 45 to be equal to the reaction that's going to be acting here, which will be acting downwards, which is equal to L by 2.
- Then you take all the forces along the x-axis, so we get the force in the member AB.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000242 · EV0000243 · EV0000244 · +1 source units in pack.json

Core explanation · 05:00–09:59

plus the force, the component of the force in the member F A D sine forty five that will be this one equal to the force that is L acting along the horizontal reaction at A.

Key details

- Then we take this joint to B.
- The force in the member B D will be zero because there is no force acting at B.
- Then the force in the member B C will be equal to the force in the member AB itself.
- Then you take joint D.
- Then we will get from joint D the force in the member B C.
- So we get the force in the member A D because here we have determined the force from joint C, then B, then we have taken D.
- But here it is joint C, then joint B and joint D.
- However you get these values.
- You get it as negative minus 0.707 L that is in this member.
- Since it is negative, it is going to be compressive.
- Then force in the member B C, we will get it as 0.5 L, which is positive, so it is tension.
- Force in the member B D will be 0 because there is no external force acting at B.
- Then force in the member AB will be the same as the force in the member B C, which is 0.5 L, which is tension.
- Force is the member, A D is also tensile, so that we will get.
- Truss would try to rotate in the clockwise direction.
- So, since this is going to rotate, then the member AD will get elongated because the force or the reaction acting here is acting downwards, and this also will be acting towards the left.
- Then when this tries to rotate along this, this member and this member would try to go down because here these two will not move, only this can move.
- Come down because of that, your AD has to elongate and AB also has to elongate.
- Both of them has to be intentional and that's what we have got here.
- I hope this you have understood.
- Of course I have explained how to determine the forces in all the members and as well as how to determine the reactions at A and C.
- And I have also told you why this truss is not a symmetrical truss.
- Moving on, I have chosen this problem, which is 4 bar 21.
- The statement of the problem goes like this.
- The signboard truss is designed to support an horizontal wind load of 4 kilometer.
- A separate analysis shows that 5 by 8 of this force is transmitted to the center connection at sea.
- And the rest is equally divided between B and B.
- Calculate the forces in member BE and BC.
- From the statement, what we can understand is there is a truss which is designed for supporting a sign mode.

- The total force that is going to be acting on the signboard because of the wind load.
- They are told what will be the portion of this total load which will be acting at each of the joints at that is at P, C and D.

Example or application from the lecture

- Wind load will be acting on an area, but they are given what is the total load that is going to be acting on this signboard as well.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000246 · EV0000247 · EV0000248 · +1 source units in pack.json

Core explanation · 10:00–14:59

And for that, there is a separate analysis that they have done, which is not included in the in this in this syllabus.

Key details

- He says that five by eight of the total load will be acting at C.
- The four that is going to be acting will be three by eight into half of that into four will be acting at D and the same load will also be acting at B.
- And also what is asked is to determine the force in the member PE and BC.
- That is force in the member BE and BC.
- Get the value of in these two members, then take joint C, get the value in these two members, then take joint B, sorry, joint E, and then get these two members.
- From this we will be able to get the value of B and B C.
- If you take the entire truss as it is, it is supported at A by a pin connection, at G also it is supported by a pin connection.
- Here also will have two reactions, one vertical and one horizontal.
- So which means that it is an indeterminate structure, we require one more equation.
- However, to determine the forces in the member BE and BC, we don't have to.
- Find out the value of the support reaction first thing.
- The second thing is.
- Since we are he has asked us to find out only the forces in B and B C, we have to solve only the joint B, joint C, and joint E only.
- We don't have to worry about solving other joints.
- How the forces is going to be acting here.
- They have explained how to get the forces in AB obviously and sigma m b equal to zero by using alternate equations of equilibrium.
- Then use the.
- Equation sigma m d equal to 0.
- Now this method that they have used is method of section.
- We will be using as I described by taking the joint D analyzing this, getting these two equations.
- or getting these value of these two, then analysing joint C, getting the value of C E and C B, and then using the joint E to get the force in the member B.

- Now the next problem that is chosen is 4 bar 23.
- Which says that if snow load transfers the forces shown to the upper joint of a prat roof truss, neglect any horizontal reactions at the supports and all for the forces in all the members.
- And if you observe all the forces are vertically acting downwards and it is equal to one kilometer.
- And we have the reaction acting at A, which is a pin connection, we have the reaction acting at E, which is also a pin connection.
- But since there is no horizontal.

Example or application from the lecture

- Let's look at what are the forces that is acting.
- Then the remaining load, that is 3 by 8, which is remaining of the four kilometers, is to be equally shared at D and C is what is given.
- Let us not worry about this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000250 · EV0000251 · EV0000252 · +1 source units in pack.json

Core explanation · 15:00–19:59

Force that is acting, we can say that the horizontal reaction at A and E will be equal to 0.

Key details

- And since the trust is symmetrical with respect to geometry as well as loading as well as the support reactions, we can say that the vertical reaction at A and E will be equal, and which will be equal to 2.5 kilonuton, which is nothing but 1 plus 1 plus 1 plus 1 plus 1, which is 5 kilonuton divided by 2.
- We get the value of alpha standing inverse of 2 by 4, and that's what is indicated here.
- Using that and considering the joint A, we get the force in the member AB under AH.
- By using the equations of equilibrium, we can get the value of AB and AH.
- Next, to solve this, first we will solve joint A as I've explained just now, then go to joint B.
- There are only two members, we can get the value in BG as well as B H.
- Then we can take the joint H, we can get the value in H G and H C.
- Then we can take the joint G.
- Where we can easily identify that the force in the member CG is zero because there is no vertical external force that is acting at G.
- Since these two is going to be equal, the force in the member CG will be zero.
- Only these four joints is what we have to solve.
- Next, you take joint B.
- What are the forces acting?
- One is external force that is acting because of the snow load one kilon.
- Then we have BH acting.
- Then we have the force in this which we have determined as 3.35 kilon tons, then the force in BC.

- We can find out what is the force in B C, which we will get as 3.35 kilonewtons.
- Then you take sigma f y equal to 0, we get the force in the member bh as 1 kilowton.
- Both of them are acting at the same angle, alpha, so these two has to be equal, so we get as three point three five.
- When I resolve this along vertical and resolve this as along vertical, they both will cancel, so we get VG as one kilometer straight away.
- Then you take the joint h and then determine what is CH and CG, resolve CH along vertical and horizontal, take sigma y equal to 0, we get ch as 1 point four one four.
- Then sigma f x equal to 0, if you take I get the force in the member gh.
- Then Cd is equal to B C, which is equal to 3.35 kilonewton, which is also compressive.
- Then EF is equal to.
- A H which is equal to 3 kilonut time which is tensile, then Bf is equal to BH which is equal to 1 kilonut time which is compressive.
- Cg is equal to CH which is equal to 1.414 kilonut time which is tension.
- Fg is equal to G H, which is equal to two kilone, which is tension, and of course CG force in the member CG is equal to zero.
- Now because of symmetry.
- We are able to get the forces in all the members by solving only three joints, and of course the joint G, if you take you get the force in the member CG.
- Because we have the same forces acting to the left of the symmetrical axis and right of the symmetrical axis.
- I am Dr.
- Steven Tesh from Civil Engineering, the Department of Science and Humanities, PS University.

Example or application from the lecture

- And that is what is given here.
- Next, consider the angle made by the line B C with the horizontal, that is angle CAG to be equal to alpha.
- Let's meet in the next class, that is class number thirty-four, and discuss more about.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000254 · EV0000255 · EV0000256 · +1 source units in pack.json

17 Source slide deck: UE26ME142A Unit 2 Class 25 Lecture PPT

Study focus: Source slide deck: UE26ME142A Unit 2 Class 25 Lecture PPT. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000515 · EV0000516

SHORT-NOTE RECAP

Study notes · Source slide deck: UE26ME142A Unit 2 Class 25 Lecture PPT

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING MECHANICS 1

- 1Engineering Mechanics-01- Unit 2 - Chapter - 4 - Method of sections

Slide 2 · CLASS - 25

- (4–0–0–4–4)
- CIVIL ENGINEERING, DEPARTMENT OF MECHANICAL
- 2Engineering Mechanics-01- Unit 2 - Chapter - 4 - Method of sections

Slide 3 · ENGINEERING MECHANICS-1-STATICS

- Structures: Introduction, Plane Trusses, Method of Joints, Numerical
- problems, Method of sections, Numerical.
- 3 Engineering Mechanics-01- Unit 2 - Chapter - 4 - Method of sections

Slide 4 · ENGINEERING MECHANICS

- 3/29, 3/36, 3/38, 3/41, 3/46, 3/50, 3/51, 3/55, 3/56. Total
- 4/13, 4/14, 4/15, 4/16, 4/18, 4/19, 4/20, 4/21, 4/22, 4/25, 4/26,
- Note: 13 Problems to be solved in class – 13 Problems 4 Engineering Mechanics-01- Unit 2 - Chapter - 4 - Method of sections

Slide 5 · ENGINEERING MECHANICS

- 5 Engineering Mechanics-01- Unit 2 - Chapter - 4 - Method of sections

Slide 6 · ENGINEERING MECHANICS - 1

- IIlustration of the method of sections
- 6 Engineering Mechanics-01- Unit 2 - Chapter - 4 - Method of sections

Slide 7 · ENGINEERING MECHANICS - 1

- Solve for unknown forces using the method of sections is:
- Determine if a truss can be modeled as a simple truss.
- 7 Engineering Mechanics-01- Unit 2 - Chapter - 4 - Method of sections•Assume that all unknown forces are in tension, then solve for

Slide 8 · Dr. S V Venkatesh

- 8Engineering Mechanics-01- Unit 2 - Chapter - 4 - Method of sections

EV0000515 · EV0000516 · EV0000517 · +21 source units in pack.json

18 Source slide deck: UE26ME142A Unit 2 Class 26 Lecture PPT

Study focus: Source slide deck: UE26ME142A Unit 2 Class 26 Lecture PPT. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000539 · EV0000540

SHORT-NOTE RECAP

Study notes · Source slide deck: UE26ME142A Unit 2 Class 26 Lecture PPT

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING MECHANICS 1

- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 1

Slide 2 · CLASS - 26

- (4–0–0–4–4)
- CIVIL ENGINEERING, DEPARTMENT OF MECHANICAL
- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 2

Slide 3 · ENGINEERING MECHANICS

- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 3

Slide 4 · ENGINEERING MECHANICS-1

- 4/31 Determine the force in member
- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 4

Slide 5 · ENGINEERING MECHANICS-1

- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 5

Slide 6 · ENGINEERING MECHANICS-1

- 4/32 Determine the force in member AE of the loaded
- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 6

Slide 7 · ENGINEERING MECHANICS-1

- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 7

Slide 8 · ENGINEERING MECHANICS-1

- 4/33 Determine the forces in members BC and
- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 8

Slide 9 · ENGINEERING MECHANICS-1

- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 9

Slide 10 · ENGINEERING MECHANICS-1

- 4/34 Determine the forces in members CG and GH of the
- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 10

Slide 11 · ENGINEERING MECHANICS-1

- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 11

Slide 12 · ENGINEERING MECHANICS-1

- 4/35 Determine the force in member BC of the loaded truss.
- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 12

Slide 13 · ENGINEERING MECHANICS-1

- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 13

Slide 14 · Dr. S V Venkatesh

- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of 14

EV0000539 · EV0000540 · EV0000541 · +39 source units in pack.json

19 Source slide deck: UE26ME142A Unit 2 Class 27 Lecture PPT

Study focus: Source slide deck: UE26ME142A Unit 2 Class 27 Lecture PPT. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000581 · EV0000582

SHORT-NOTE RECAP

Study notes · Source slide deck: UE26ME142A Unit 2 Class 27 Lecture PPT

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING MECHANICS 1

- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of

Slide 2 · CLASS - 27

- (4–0–0–4–4)
- CIVIL ENGINEERING, DEPARTMENT OF MECHANICAL
- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of

Slide 3 · ENGINEERING MECHANICS-1

- 4/36 Determine the force in member BE of the loaded truss.
- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of

Slide 4 · ENGINEERING MECHANICS-1

- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of

Slide 5 · ENGINEERING MECHANICS-1

- 4/37 Determine the forces in members DE and DL.
- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of

Slide 6 · ENGINEERING MECHANICS-1

- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of

Slide 7 · ENGINEERING MECHANICS-1

- 4/38 Calculate the forces in members BC, BE, and EF. Solve for
- each force from an equilibrium equation which contains that
- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of

Slide 8 · ENGINEERING MECHANICS-1

- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of

Slide 9 · ENGINEERING MECHANICS-1

- 4/39 Determine the forces in members BC and CF of the loaded
- truss, repeated here from Prob. 4/19.
- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of

Slide 10 · ENGINEERING MECHANICS-1

- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of

Slide 11 · ENGINEERING MECHANICS-1

- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of

Slide 12 · ENGINEERING MECHANICS-1

- 4/40 Calculate the forces in members BC, CD, and CG of the

- loaded truss composed of equilateral triangles, each of side
- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of

Slide 13 · ENGINEERING MECHANICS-1

- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of

Slide 14 · ENGINEERING MECHANICS-1

- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of

Slide 15 · Dr. S V Venkatesh

- Engineering Mechanics-01 - Unit 2 - Chapter - 4 - Method of

EV0000581 · EV0000582 · EV0000583 · +42 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
UE26ME142A_Unit 2_Class 14_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/014_UE26ME142A_Unit_2_Class_14_AVSummary__d028b1de67591979.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 14_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/014_UE26ME142A_Unit_2_Class_14_AVSummary__d028b1de67591979.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 14_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/014_UE26ME142A_Unit_2_Class_14_AVSummary__d028b1de67591979.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 14_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/014_UE26ME142A_Unit_2_Class_14_AVSummary__d028b1de67591979.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 15_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/015_UE26ME142A_Unit_2_Class_15_AVSummary__29e4398836d93577.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 15_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/015_UE26ME142A_Unit_2_Class_15_AVSummary__29e4398836d93577.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 15_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/015_UE26ME142A_Unit_2_Class_15_AVSummary__29e4398836d93577.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 15_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/015_UE26ME142A_Unit_2_Class_15_AVSummary__29e4398836d93577.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 16_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/016_UE26ME142A_Unit_2_Class_16_AVSummary__12354184f0f96c13.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 16_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/016_UE26ME142A_Unit_2_Class_16_AVSummary__12354184f0f96c13.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 16_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/016_UE26ME142A_Unit_2_Class_16_AVSummary__12354184f0f96c13.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9996; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 16_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/016_UE26ME142A_Unit_2_Class_16_AVSummary__12354184f0f96c13.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 17_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/017_UE26ME142A_Unit_2_Class_17_AVSummary__25ac1a3b8349a649.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 17_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/017_UE26ME142A_Unit_2_Class_17_AVSummary__25ac1a3b8349a649.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 17_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/017_UE26ME142A_Unit_2_Class_17_AVSummary__25ac1a3b8349a649.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 17_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/017_UE26ME142A_Unit_2_Class_17_AVSummary__25ac1a3b8349a649.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 18_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/018_UE26ME142A_Unit_2_Class_18_AVSummary__dafbae985b8a08a6.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 18_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/018_UE26ME142A_Unit_2_Class_18_AVSummary__dafbae985b8a08a6.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 18_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/018_UE26ME142A_Unit_2_Class_18_AVSummary__dafbae985b8a08a6.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 18_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/018_UE26ME142A_Unit_2_Class_18_AVSummary__dafbae985b8a08a6.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 19.0_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.0_AVSummary__bd8cd9929f8bd715.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.0_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.0_AVSummary__bd8cd9929f8bd715.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.0_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.0_AVSummary__bd8cd9929f8bd715.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.0_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.0_AVSummary__bd8cd9929f8bd715.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.1_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.1_AVSummary__877d29f4ad0c369d.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.1_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.1_AVSummary__877d29f4ad0c369d.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 19.1_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.1_AVSummary__877d29f4ad0c369d.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.1_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.1_AVSummary__877d29f4ad0c369d.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.2_AVSummary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.2_AVSummary__ee992a99bb75e05b.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.2_AVSummary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.2_AVSummary__ee992a99bb75e05b.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.2_AVSummary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.2_AVSummary__ee992a99bb75e05b.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9992; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.2_AVSummary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.2_AVSummary__ee992a99bb75e05b.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 19.3_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.3_AV_Summary__d60756405f7bf7e2.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.3_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.3_AV_Summary__d60756405f7bf7e2.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.3_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.3_AV_Summary__d60756405f7bf7e2.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.3_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.3_AV_Summary__d60756405f7bf7e2.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.4_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.4_AV_Summary__5e4cbdbb6d761857.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.4_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.4_AV_Summary__5e4cbdbb6d761857.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 19.4_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.4_AV_Summary__5e4cbdbb6d761857.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.4_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.4_AV_Summary__5e4cbdbb6d761857.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.5_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.5_AV_Summary__145da9b4503db2ea.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.5_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.5_AV_Summary__145da9b4503db2ea.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.5_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.5_AV_Summary__145da9b4503db2ea.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.5_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.5_AV_Summary__145da9b4503db2ea.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 21_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/021_UE26ME142A_Unit_2_Class_21_AV_Summary__e2f3aefd30dcda28.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 21_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/021_UE26ME142A_Unit_2_Class_21_AV_Summary__e2f3aefd30dcda28.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 21_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/021_UE26ME142A_Unit_2_Class_21_AV_Summary__e2f3aefd30dcda28.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 21_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/021_UE26ME142A_Unit_2_Class_21_AV_Summary__e2f3aefd30dcda28.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 22_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/022_UE26ME142A_Unit_2_Class_22_AV_Summary__ebcc1256b1d0786a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 22_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/022_UE26ME142A_Unit_2_Class_22_AV_Summary__ebcc1256b1d0786a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 22_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/022_UE26ME142A_Unit_2_Class_22_AV_Summary__ebcc1256b1d0786a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 22_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/022_UE26ME142A_Unit_2_Class_22_AV_Summary__ebcc1256b1d0786a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 23_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/023_UE26ME142A_Unit_2_Class_23_AV_Summary__d2db844898691c1a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 23_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/023_UE26ME142A_Unit_2_Class_23_AV_Summary__d2db844898691c1a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 23_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/023_UE26ME142A_Unit_2_Class_23_AV_Summary__d2db844898691c1a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 23_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/023_UE26ME142A_Unit_2_Class_23_AV_Summary__d2db844898691c1a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 24_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/024_UE26ME142A_Unit_2_Class_24_AV_Summary__ddeae52719c2a16c.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 24_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/024_UE26ME142A_Unit_2_Class_24_AV_Summary__ddeae52719c2a16c.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 24_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/024_UE26ME142A_Unit_2_Class_24_AV_Summary__ddeae52719c2a16c.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 24_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/024_UE26ME142A_Unit_2_Class_24_AV_Summary__ddeae52719c2a16c.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: UE26ME142A Unit 2 Class 14 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 14 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 2 Class 14 Lecture PPT (converted PDF; original hash 15da602433d67b8f691b145df45757178b9b3848800a2a73732c9738e0f13ae4)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 15 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 15 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 2 Class 15 Lecture PPT (converted PDF; original hash 89ba5293343d1098702f71e08aad13f40b1746763b94a71a51431b22641f12f0)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 16 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 16 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 2 Class 16 Lecture PPT (converted PDF; original hash b92d5b6449d2b1d0e7939ef64985be8ceee8a6dcc6e9b2b4d9c54b8e8cda12cf)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 17 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 17 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 2 Class 17 Lecture PPT (converted PDF; original hash 17bd7c62545d7149617f64fd0d4cf7a538e17facccb990bc698a31ce012ad248)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 18 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 18 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

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Page-render source for UE26ME142A Unit 2 Class 18 Lecture PPT (converted PDF; original hash 85c8ad866d36397419056cc6aa0adf9538c8032a3f4d265f9ff53ea40cb9c317)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 19.0 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 19.0 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 2 Class 19.0 Lecture PPT (converted PDF; original hash a8da7337867f3a999244592eb71afafb59434ce313cf1b11c4d6f5f59348ed93)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 19.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 19.1 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

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Page-render source for UE26ME142A Unit 2 Class 19.1 Lecture PPT (converted PDF; original hash 4688882de86918464ab39ce7074721d5300631f00b354558a4dd8f850a28bbeb)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 19.2 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 19.2 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 2 Class 19.2 Lecture PPT (converted PDF; original hash a5ba8cda3b81747db031cb68ac3d86e434f1f076393e4f47c1c1b1aebe018e0c)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 19.3 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 19.3 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 2 Class 19.3 Lecture PPT (converted PDF; original hash 9c0be305589bdb3099cdb1dd33474c3d8487f218b47a12992ccc419a2fd76e22)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 19.4 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 19.4 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 2 Class 19.4 Lecture PPT (converted PDF; original hash 6ee58b950fa60fb1cf9861566536c6dfcde27c27b3b7d9765b64ffdabd263eef)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 19.5 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 19.5 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

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Page-render source for UE26ME142A Unit 2 Class 19.5 Lecture PPT (converted PDF; original hash c4c0b2c1da6ec17d1df4917d1457210a2f14451595cb0834e64b921b6245c826)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 21 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 21 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 2 Class 21 Lecture PPT (converted PDF; original hash 654ff6d174c162369b8262db13c6dfdebeb6e04b9ef5ff4d0e59bba981cc4734)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 22 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 22 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 2 Class 22 Lecture PPT (converted PDF; original hash 94a72dcc07893b420ffe9ef97557283604675f4a32e93c57b79637e014454b0c)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 23 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
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Page-render source for UE26ME142A Unit 2 Class 23 Lecture PPT (converted PDF; original hash 416e21f70149398de40b961e40fe20526efc7233cc456d68b7461737cfb9ee15)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 24 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 24 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 2 Class 24 Lecture PPT (converted PDF; original hash 7f455b5442d047c16abcc290ac99dc476657492e34a722aeb825b1fbc964f6c2)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 25 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 25 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 2 Class 25 Lecture PPT (converted PDF; original hash 3f7038dca9063fae74f933716599fb7871a17fcb789a660a382bf2632613532c)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/025_UE26ME142A_Unit_2_Class_25_Lecture_PPT__9ba1358e9a09f7e6.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 26 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 26 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE26ME142A Unit 2 Class 26 Lecture PPT (converted PDF; original hash e912704c214b466e06ae34f25b61ac6dbfc415fa8d05d19c2148b6b9c5e9a360)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/026_UE26ME142A_Unit_2_Class_26_Lecture_PPT__7d399883b2389a11.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 2 Class 27 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 2 Class 27 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 2 Class 27 Lecture PPT (converted PDF; original hash 54f8351c620edee63d6f52420621272348097c94fa3bf919fd13226ed2925b29)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/Slides/027_UE26ME142A_Unit_2_Class_27_Lecture_PPT__e9ae146210fa8e2e.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md have no readable `.slides.md` text; their original page images are included.
2. 4 rendered pages in 015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.slides.md have no readable `.slides.md` text; their original page images are included.

4. 4 rendered pages in 017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.slides.md have no readable `.slides.md` text; their original page images are included.
7. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.slides.md have no readable `.slides.md` text; their original page images are included.
9. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.slides.md have no readable `.slides.md` text; their original page images are included.
10. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.slides.md have no readable `.slides.md` text; their original page images are included.
12. 3 rendered pages in 021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.slides.md have no readable `.slides.md` text; their original page images are included.
14. 3 rendered pages in 023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.slides.md have no readable `.slides.md` text; their original page images are included.
16. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
17. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.

18. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
19. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
20. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00001	1 covered · 0 excluded · 0 gaps	catalogue: Unit title and index copied from the official units.json catalogue; human content review is pending.
SRC00002	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00003	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...
SRC00004	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00005	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare...
SRC00006	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00007	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...
SRC00008	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00009	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0053. Sidecar variants compare...
SRC00010	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00011	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare...
SRC00012	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00013	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare...
SRC00014	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00015	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...
SRC00016	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00017	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...
SRC00018	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00019	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare...
SRC00020	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00021	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare...
SRC00022	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0056. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00023	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0056. Sidecar variants compare...
SRC00024	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0056. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00025	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0056. Sidecar variants compare...
SRC00026	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0034. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0035 to s0050. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00027	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0034. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0035 to s0050. Sidecar variants compare...
SRC00028	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0034. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0035 to s0050. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00029	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0034. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0035 to s0050. Sidecar variants compare...
SRC00030	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00031	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare...
SRC00032	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00033	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0054. Sidecar variants compare...
SRC00034	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00035	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...
SRC00036	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00037	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...
SRC00038	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00039	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare...
SRC00040	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00041	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0018 to s0036. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0037 to s0053. Sidecar variants compare...
SRC00042	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00043	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...
SRC00044	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00045	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...
SRC00046	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00047	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...
SRC00048	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00049	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare...
SRC00050	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00051	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...
SRC00052	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00053	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...
SRC00054	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00055	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...
SRC00056	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00057	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...
SRC00058	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00059	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00060	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00061	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00062	0 covered · 0 excluded · 8 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00064	0 covered · 0 excluded · 10 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00063	0 covered · 0 excluded · 8 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00065	0 covered · 0 excluded · 3 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00067	0 covered · 0 excluded · 7 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00066	0 covered · 0 excluded · 3 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; text differs from or is absent in the .slides.md page;...
SRC00068	0 covered · 0 excluded · 5 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00070	0 covered · 0 excluded · 7 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00069	0 covered · 0 excluded · 5 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00071	0 covered · 0 excluded · 2 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from...
SRC00073	0 covered · 0 excluded · 6 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00072	0 covered · 0 excluded · 2 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human...
SRC00074	0 covered · 0 excluded · 3 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...

Source	Unit counts	Example decisions
SRC00076	0 covered · 0 excluded · 6 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00075	0 covered · 0 excluded · 3 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00077	0 covered · 0 excluded · 3 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00079	0 covered · 0 excluded · 6 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00078	0 covered · 0 excluded · 3 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; text differs from or is absent in the .slides.md page;...
SRC00080	0 covered · 0 excluded · 4 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00082	0 covered · 0 excluded · 6 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00081	0 covered · 0 excluded · 4 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00083	0 covered · 0 excluded · 2 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from...

Source	Unit counts	Example decisions
SRC00085	0 covered · 0 excluded · 6 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00084	0 covered · 0 excluded · 2 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human...
SRC00086	0 covered · 0 excluded · 4 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00088	0 covered · 0 excluded · 7 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00087	0 covered · 0 excluded · 4 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; text differs from or is absent in the .slides.md page;...

Source	Unit counts	Example decisions
SRC00089	0 covered · 0 excluded · 6 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00091	0 covered · 0 excluded · 8 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00090	0 covered · 0 excluded · 6 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00092	0 covered · 0 excluded · 4 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00094	0 covered · 0 excluded · 6 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00093	0 covered · 0 excluded · 4 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00095	0 covered · 0 excluded · 15 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00097	0 covered · 0 excluded · 18 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00096	0 covered · 0 excluded · 15 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00098	0 covered · 0 excluded · 3 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...

Source	Unit counts	Example decisions
SRC00100	0 covered · 0 excluded · 5 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00099	0 covered · 0 excluded · 3 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00101	0 covered · 0 excluded · 5 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00103	0 covered · 0 excluded · 8 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00102	0 covered · 0 excluded · 5 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00104	0 covered · 0 excluded · 5 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00106	0 covered · 0 excluded · 7 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00105	0 covered · 0 excluded · 5 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00107	8 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...

Source	Unit counts	Example decisions
SRC00109	8 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00108	8 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00110	14 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00112	14 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00111	14 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00113	15 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00115	15 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00114	15 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

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